

Final Report

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Introduction

The data set I have chosen is ‘Student Performance DataSet’. I found this data interesting because I get to find out what all factors that are affecting the outcome of a student performance by performing data analysis. The most basic things which I am curious about is what makes a student switch schools, how does the time a student put in work affect their grades, how does their relations to their family members affect their health and how much does having an educational support affect the student’s performance. This kind of data analysis helps the school understand what works for students and what doesn’t which allows them to better their methods to get better results.

Data

Data Source

Citation: P. Cortez and A. Silva. Using Data Mining to Predict Secondary School Student Performance. In A. Brito and J. Teixeira Eds., Proceedings of 5th FUTURE BUSINESS TECHNOLOGY Conference (FUBUTEC 2008) pp. 5-12, Porto, Portugal, April, 2008, EUROSIS, ISBN 978-9077381-39-7.

Link: <http://archive.ics.uci.edu/ml/datasets/Student+Performance>

Data Collection

This data approaches student achievement in secondary education of two Portuguese schools. It was collected in 2005-2006 school year. The data was collected mainly from school reports and additional questionnaires.

Units of observation

Each row of this data indicates a record for a particular student.

Variables

The following are the variables in the dataset which I will be studying.

```
library(knitr)
library(kableExtra)
variable <- read.csv(file = 'data/variable_details.csv')
kbl(variable, booktabs=T)
```

S.No.	Variable	Details
1	school	student's school (binary: 'GP' - Gabriel Pereira or 'MS' - Mousinho da Silveira)
2	sex	student's sex (binary: 'F' - female or 'M' - male)
3	age	student's age (numeric: from 15 to 22)
4	address	student's home address type (binary: 'U' - urban or 'R' - rural)
5	famsize	family size (binary: 'LE3' - less or equal to 3 or 'GT3' - greater than 3)
6	Pstatus	parent's cohabitation status (binary: 'T' - living together or 'A' - apart)
7	Medu	mother's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education)
8	Fedu	father's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education)
9	Mjob	mother's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'other'))
10	Fjob	father's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'other'))
11	reason	reason to choose this school (nominal: close to 'home', school 'reputation', 'course' preference or 'other'))
12	guardian	student's guardian (nominal: 'mother', 'father' or 'other')
13	traveltime	home to school travel time (numeric: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)
14	studytime	weekly study time (numeric: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours)
15	failures	number of past class failures (numeric: n if 1<=n<3, else 4)
16	schoolsup	extra educational support (binary: yes or no)
17	famsup	family educational support (binary: yes or no)
18	paid	extra paid classes within the course subject (Math or Portuguese) (binary: yes or no)
19	activities	extra-curricular activities (binary: yes or no)
20	nursery	attended nursery school (binary: yes or no)
21	higher	wants to take higher education (binary: yes or no)
22	internet	Internet access at home (binary: yes or no)
23	romantic	with a romantic relationship (binary: yes or no)
24	famrel	quality of family relationships (numeric: from 1 - very bad to 5 - excellent)
25	freetime	free time after school (numeric: from 1 - very low to 5 - very high)
26	goout	going out with friends (numeric: from 1 - very low to 5 - very high)
27	Dalc	workday alcohol consumption (numeric: from 1 - very low to 5 - very high)
28	Walc	weekend alcohol consumption (numeric: from 1 - very low to 5 - very high)
29	health	current health status (numeric: from 1 - very bad to 5 - very good)
30	absences	number of school absences (numeric: from 0 to 93)
31	G1	first period grade (numeric: from 0 to 20)
32	G2	second period grade (numeric: from 0 to 20)
33	G3	final grade (numeric: from 0 to 20, output target)

Type of Study

This was an observational study.

Data Clean-Up

No data cleaning was performed. There were two datasets present, one for Mathematics subject and another for Portuguese subject. We will be using the Portuguese subject dataset.

Exploratory Data Analysis

```
library(tidyverse)
```

```
## -- Attaching packages ----- tidyverse 1.3.1 --
```

```
## v ggplot2 3.3.5      v purrr 0.3.4
## v tibble 3.1.5       v dplyr 1.0.7
## v tidyr 1.1.3        v stringr 1.4.0
## v readr 2.0.2        v forcats 0.5.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter()      masks stats::filter()
## x dplyr::group_rows() masks kableExtra::group_rows()
## x dplyr::lag()         masks stats::lag()
```

```
student <- read.csv2(file = 'data/student-por.csv')
head(student)
```

```
##   school sex age address famsize Pstatus Medu Fedu   Mjob   Fjob   reason
## 1    GP   F  18      U    GT3      A    4    4  at_home teacher  course
## 2    GP   F  17      U    GT3      T    1    1  at_home  other  course
## 3    GP   F  15      U    LE3      T    1    1  at_home  other  other
## 4    GP   F  15      U    GT3      T    4    2  health services  home
## 5    GP   F  16      U    GT3      T    3    3   other  other  home
## 6    GP   M  16      U    LE3      T    4    3 services  other reputation
##   guardian traveltime studytime failures schoolsup famsup paid activities
## 1  mother          2          2          0        yes    no  no          no
## 2  father          1          2          0        no    yes  no          no
## 3  mother          1          2          0        yes    no  no          no
## 4  mother          1          3          0        no    yes  no          yes
## 5  father          1          2          0        no    yes  no          no
## 6  mother          1          2          0        no    yes  no          yes
##   nursery higher internet romantic famrel freetime goout Dalc Walc health
## 1    yes    yes      no      no      4          3    4    1    1    3
## 2     no    yes      yes      no      5          3    3    1    1    3
## 3    yes    yes      yes      no      4          3    2    2    3    3
## 4    yes    yes      yes      yes      3          2    2    1    1    5
## 5    yes    yes      no      no      4          3    2    1    2    5
## 6    yes    yes      yes      no      5          4    2    1    2    5
##   absences G1 G2 G3
## 1      4  0 11 11
## 2      2  9 11 11
## 3      6 12 13 12
## 4      0 14 14 14
## 5      0 11 13 13
## 6      6 12 12 13
```

```
student <- within(student, {
  studytime.cat <- NA
  studytime.cat[studytime == 1] <- "Less than 2 hours"
  studytime.cat[studytime == 2] <- "2-5 hours"
  studytime.cat[studytime == 3] <- "5-10 hours"
  studytime.cat[studytime == 4] <- "More than 10 hours"
```

```

famrel.cat <- NA
famrel.cat[famrel == 1] <- "Very Bad"
famrel.cat[famrel == 2] <- "Bad"
famrel.cat[famrel == 3] <- "Good"
famrel.cat[famrel == 4] <- "Very Good"
famrel.cat[famrel == 5] <- "Excellent"

Dalc.cat <- NA
Dalc.cat[Dalc == 1] <- "Very Low"
Dalc.cat[Dalc == 2] <- "Low"
Dalc.cat[Dalc == 3] <- "Normal"
Dalc.cat[Dalc == 4] <- "High"
Dalc.cat[Dalc == 5] <- "Very High"

health.cat <- NA
health.cat[health == 1] <- "Very Bad"
health.cat[health == 2] <- "Bad"
health.cat[health == 3] <- "Normal"
health.cat[health == 4] <- "Good"
health.cat[health == 5] <- "Very Good"

goout.cat <- NA
goout.cat[goout == 1] <- "Very Low"
goout.cat[goout == 2] <- "Low"
goout.cat[goout == 3] <- "Normal"
goout.cat[goout == 4] <- "High"
goout.cat[goout == 5] <- "Very High"

failures.cat <- NA
failures.cat[failures == 0] <- "0"
failures.cat[failures == 1] <- "1"
failures.cat[failures == 2] <- "2"
failures.cat[failures == 3] <- "3"
failures.cat[failures == 4] <- "4 and more"
} )

summary(student)

```

Visualizations

##	school	sex	age	address
##	Length:649	Length:649	Min. :15.00	Length:649
##	Class :character	Class :character	1st Qu.:16.00	Class :character
##	Mode :character	Mode :character	Median :17.00	Mode :character
##			Mean :16.74	
##			3rd Qu.:18.00	
##			Max. :22.00	
##	famsize	Pstatus	Medu	Fedu
##	Length:649	Length:649	Min. :0.000	Min. :0.000
##	Class :character	Class :character	1st Qu.:2.000	1st Qu.:1.000
##	Mode :character	Mode :character	Median :2.000	Median :2.000
##			Mean :2.515	Mean :2.307

```

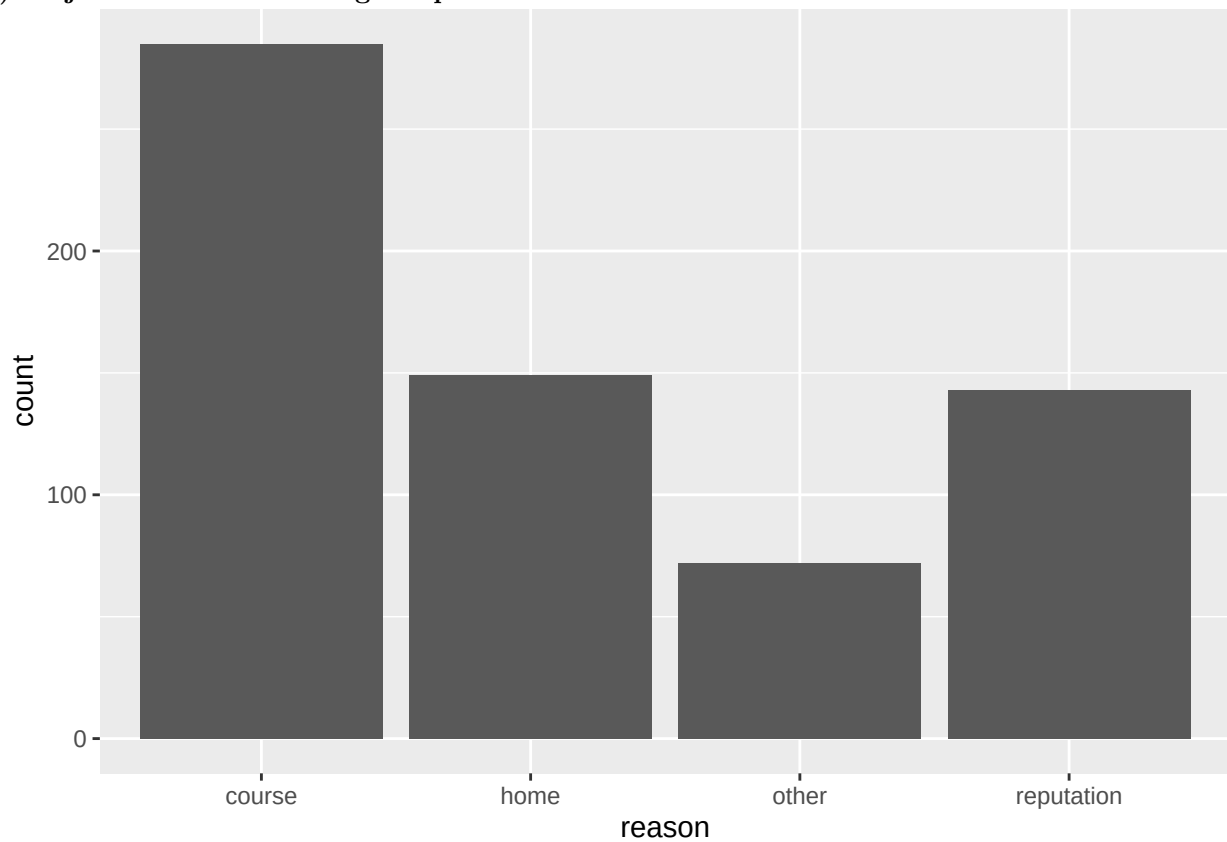
##                                     3rd Qu.:4.000  3rd Qu.:3.000
##                                     Max.    :4.000  Max.    :4.000
##      Mjob          Fjob          reason          guardian
## Length:649      Length:649      Length:649      Length:649
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##      traveltime      studytime      failures      schoolsup
## Min.    :1.000      Min.    :1.000      Min.    :0.0000      Length:649
## 1st Qu.:1.000      1st Qu.:1.000      1st Qu.:0.0000      Class :character
## Median :1.000      Median :2.000      Median :0.0000      Mode  :character
## Mean   :1.569      Mean   :1.931      Mean   :0.2219
## 3rd Qu.:2.000      3rd Qu.:2.000      3rd Qu.:0.0000
## Max.   :4.000      Max.   :4.000      Max.   :3.0000
##      famsup          paid          activities      nursery
## Length:649      Length:649      Length:649      Length:649
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##      higher          internet          romantic          famrel
## Length:649      Length:649      Length:649      Min.    :1.000
## Class :character Class :character Class :character 1st Qu.:4.000
## Mode  :character Mode  :character Mode  :character Median :4.000
##                                     Mean   :3.931
##                                     3rd Qu.:5.000
##                                     Max.   :5.000
##      freetime      goout          Dalc          Walc          health
## Min.    :1.00      Min.    :1.000      Min.    :1.000      Min.    :1.00      Min.    :1.000
## 1st Qu.:3.00      1st Qu.:2.000      1st Qu.:1.000      1st Qu.:1.00      1st Qu.:2.000
## Median :3.00      Median :3.000      Median :1.000      Median :2.00      Median :4.000
## Mean   :3.18      Mean   :3.185      Mean   :1.502      Mean   :2.28      Mean   :3.536
## 3rd Qu.:4.00      3rd Qu.:4.000      3rd Qu.:2.000      3rd Qu.:3.00      3rd Qu.:5.000
## Max.   :5.00      Max.   :5.000      Max.   :5.000      Max.   :5.00      Max.   :5.000
##      absences      G1          G2          G3
## Min.    : 0.000      Min.    : 0.0      Min.    : 0.00      Min.    : 0.00
## 1st Qu.: 0.000      1st Qu.:10.0      1st Qu.:10.00      1st Qu.:10.00
## Median : 2.000      Median :11.0      Median :11.00      Median :12.00
## Mean   : 3.659      Mean   :11.4      Mean   :11.57      Mean   :11.91
## 3rd Qu.: 6.000      3rd Qu.:13.0      3rd Qu.:13.00      3rd Qu.:14.00
## Max.   :32.000      Max.   :19.0      Max.   :19.00      Max.   :19.00
##      failures.cat      goout.cat      health.cat      Dalc.cat
## Length:649      Length:649      Length:649      Length:649
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##      famrel.cat      studytime.cat
## Length:649      Length:649
## Class :character Class :character

```

```
## Mode :character Mode :character
##
##
##
```

```
ggplot(student) +
  geom_bar(mapping = aes(x=reason))
```

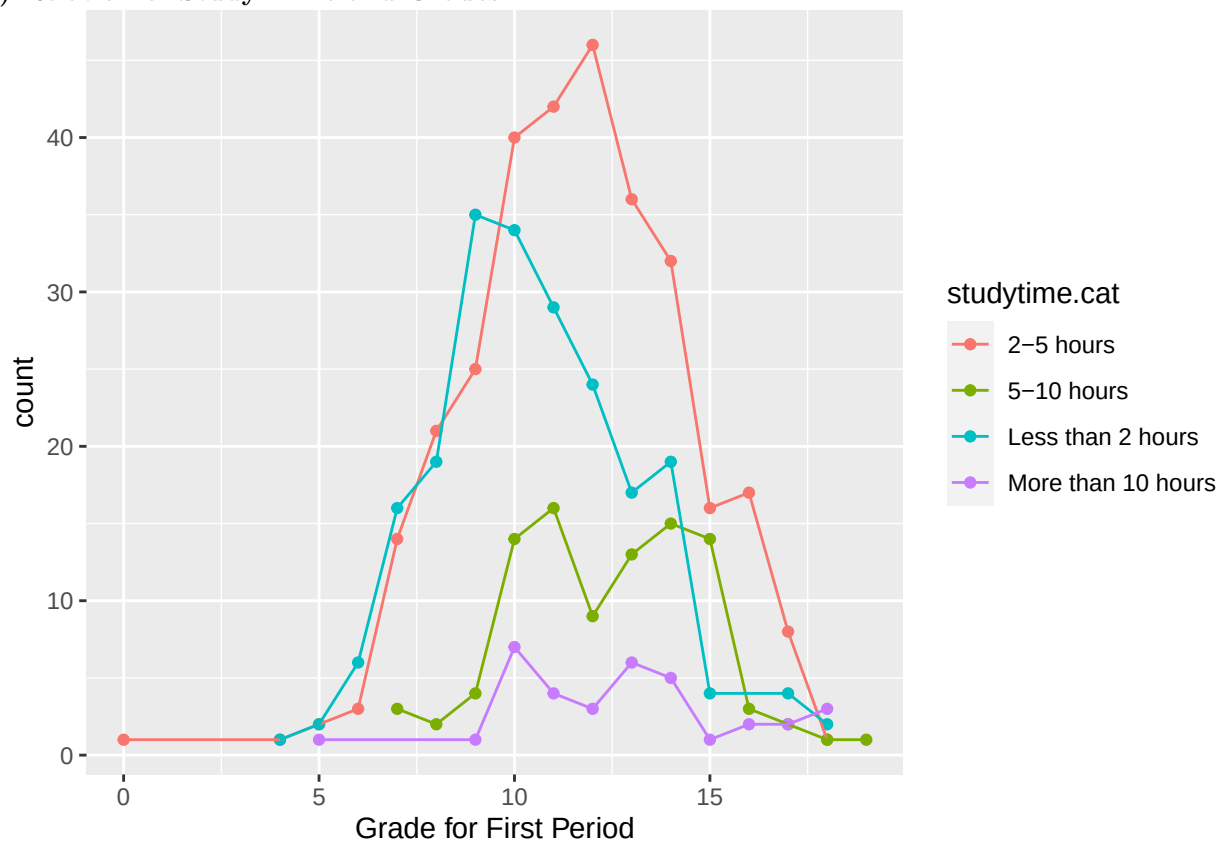
1) Major reason for choosing the particular school



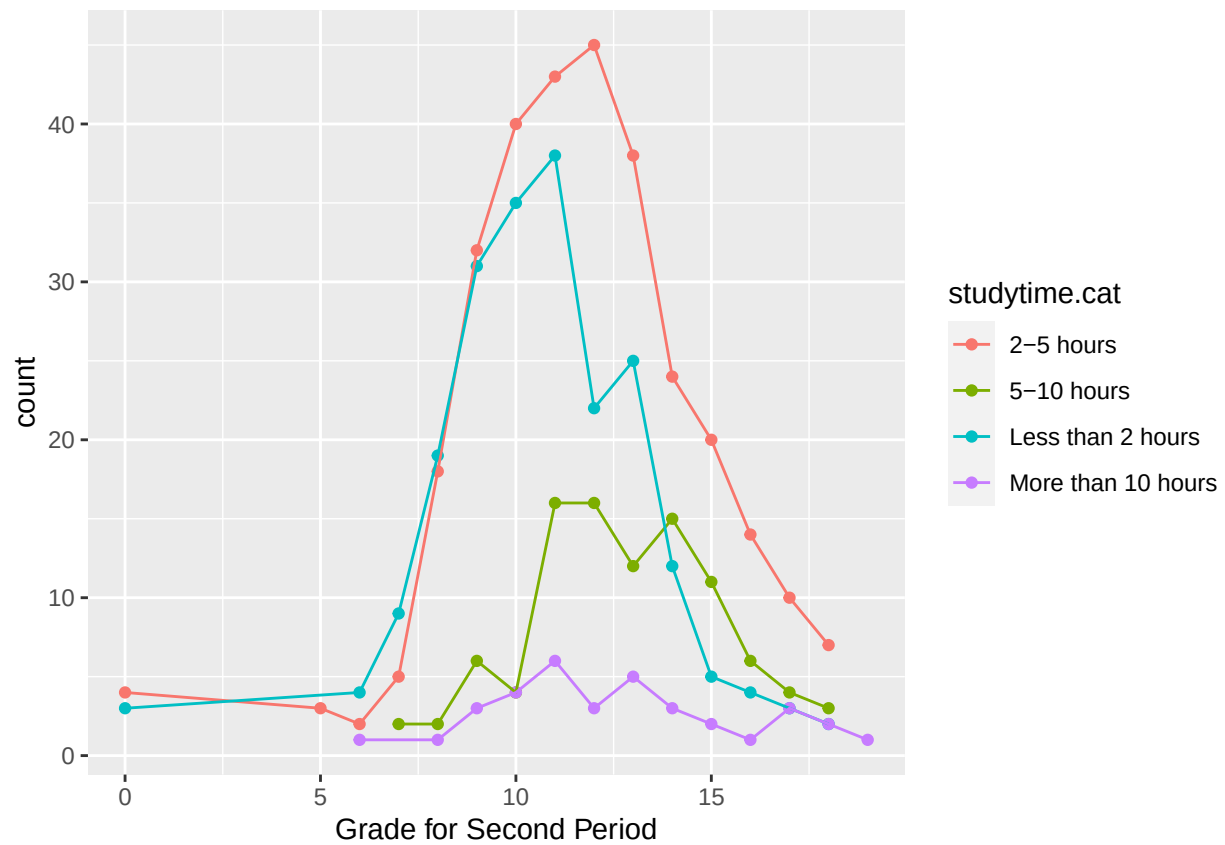
From the visualization, we can say that main reason for choosing the school was the course curriculum it provided.

```
ggplot(data = student, mapping = aes(x=G1 , colour=studytime.cat)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  xlab("Grade for First Period")
```

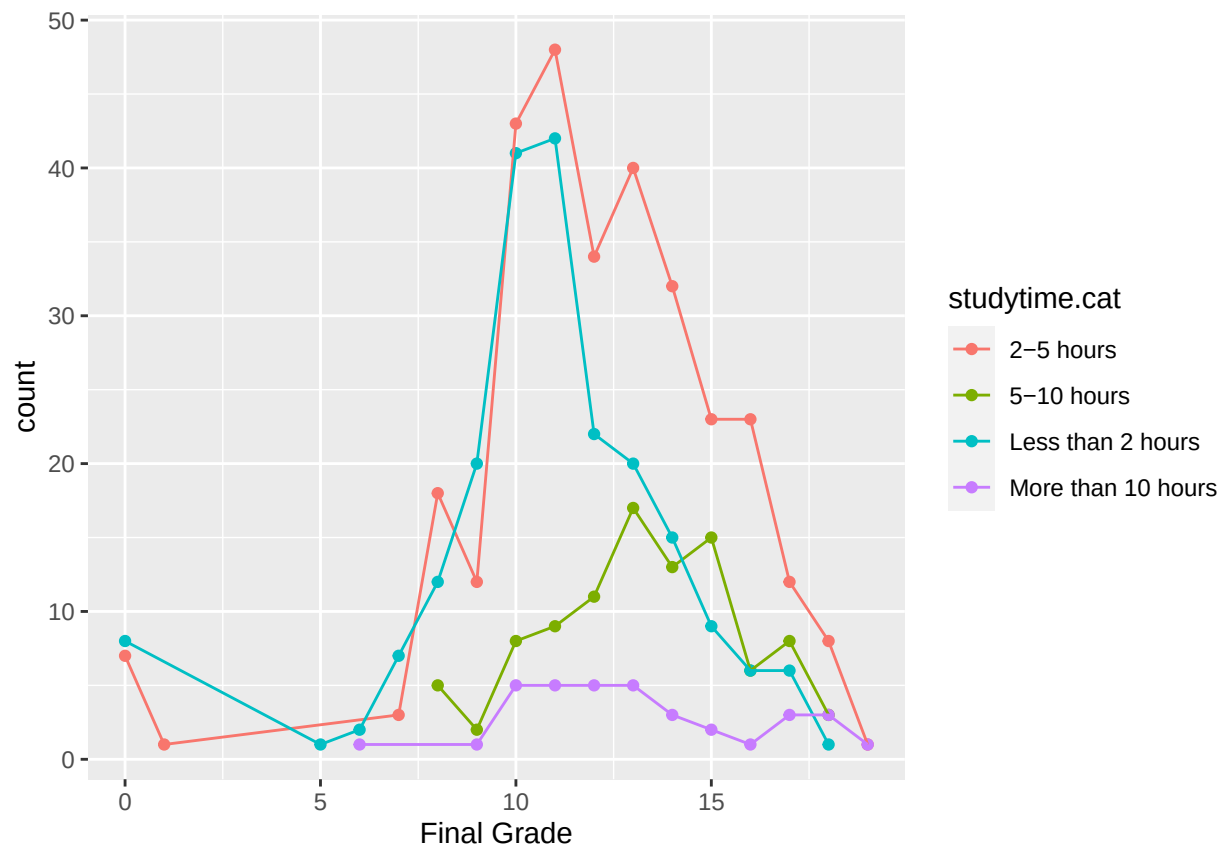
2) Relation of Study Time and Grades



```
ggplot(data = student, mapping = aes(x=G2, colour=studytime.cat)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  xlab("Grade for Second Period")
```



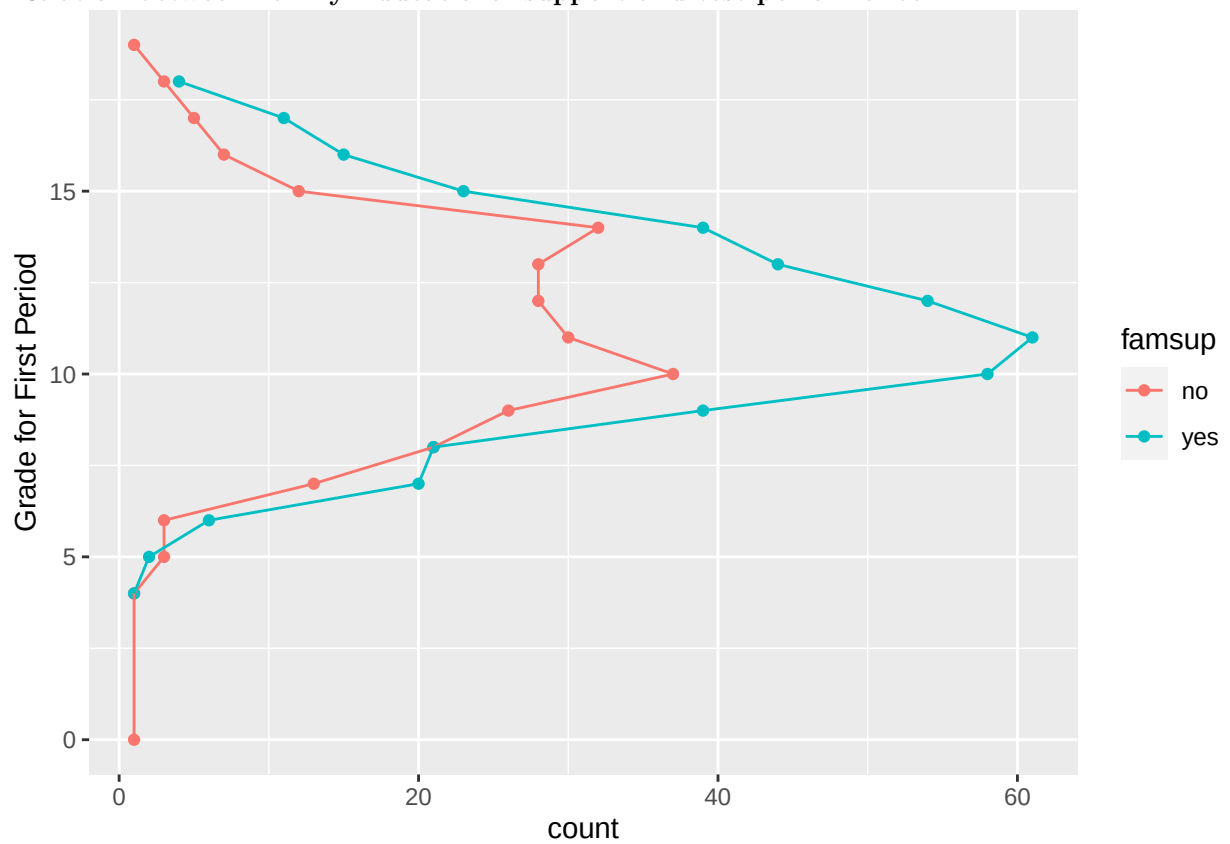
```
ggplot(data = student, mapping = aes(x=G3, colour=studytime.cat)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  xlab("Final Grade")
```



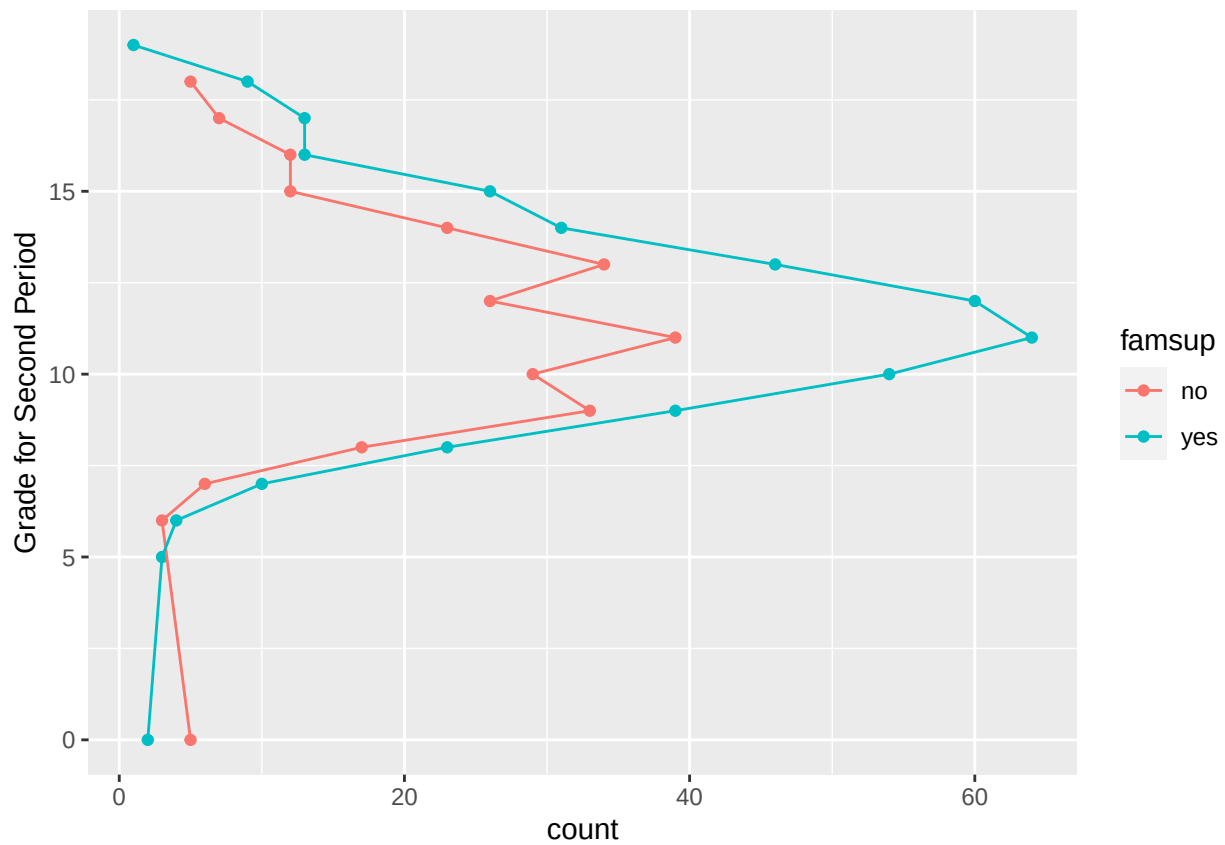
From the visualization, we can say that there are a very few students studying for a long period of time and getting bad grades.

```
ggplot(data = student, mapping = aes(y=G1, colour=famsup)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for First Period")
```

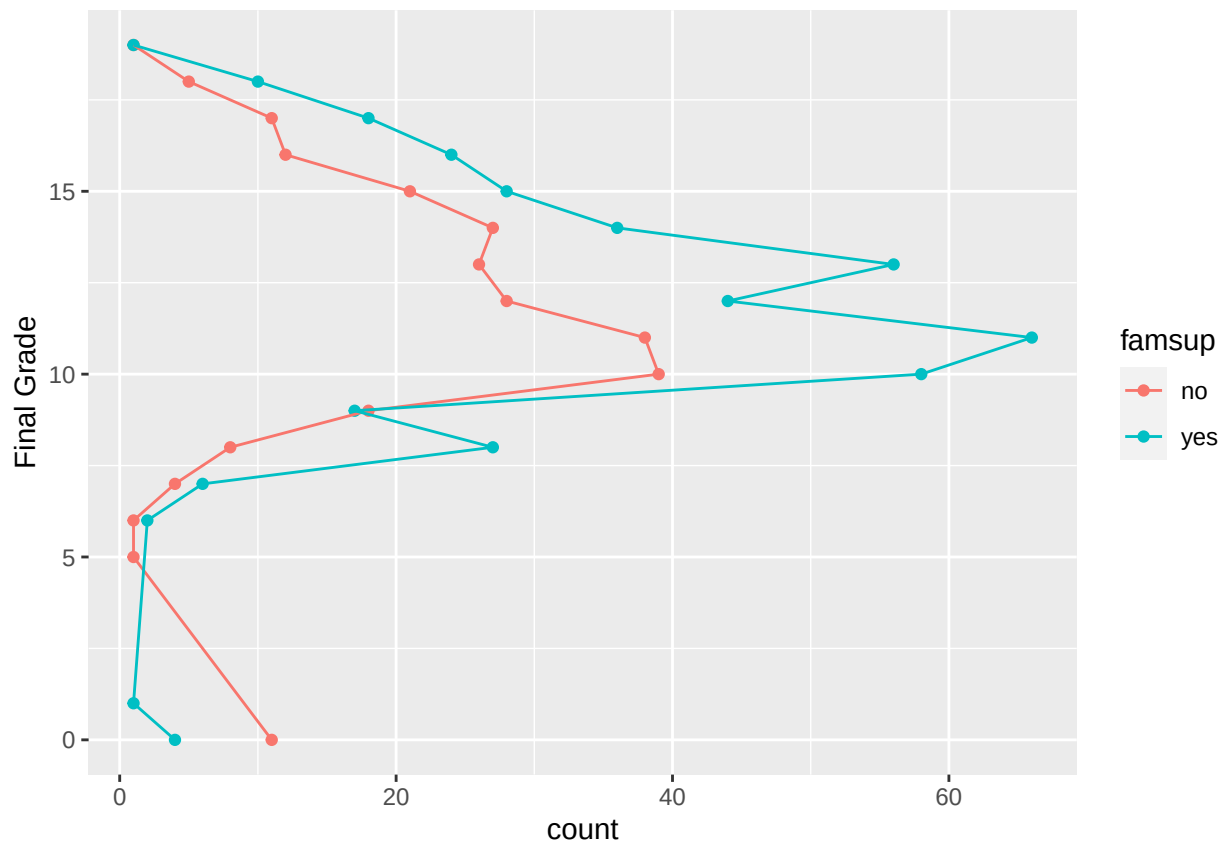
3) Relation between Family Educational support and test performance



```
ggplot(data = student, mapping = aes(y=G2, colour=famsup)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for Second Period")
```

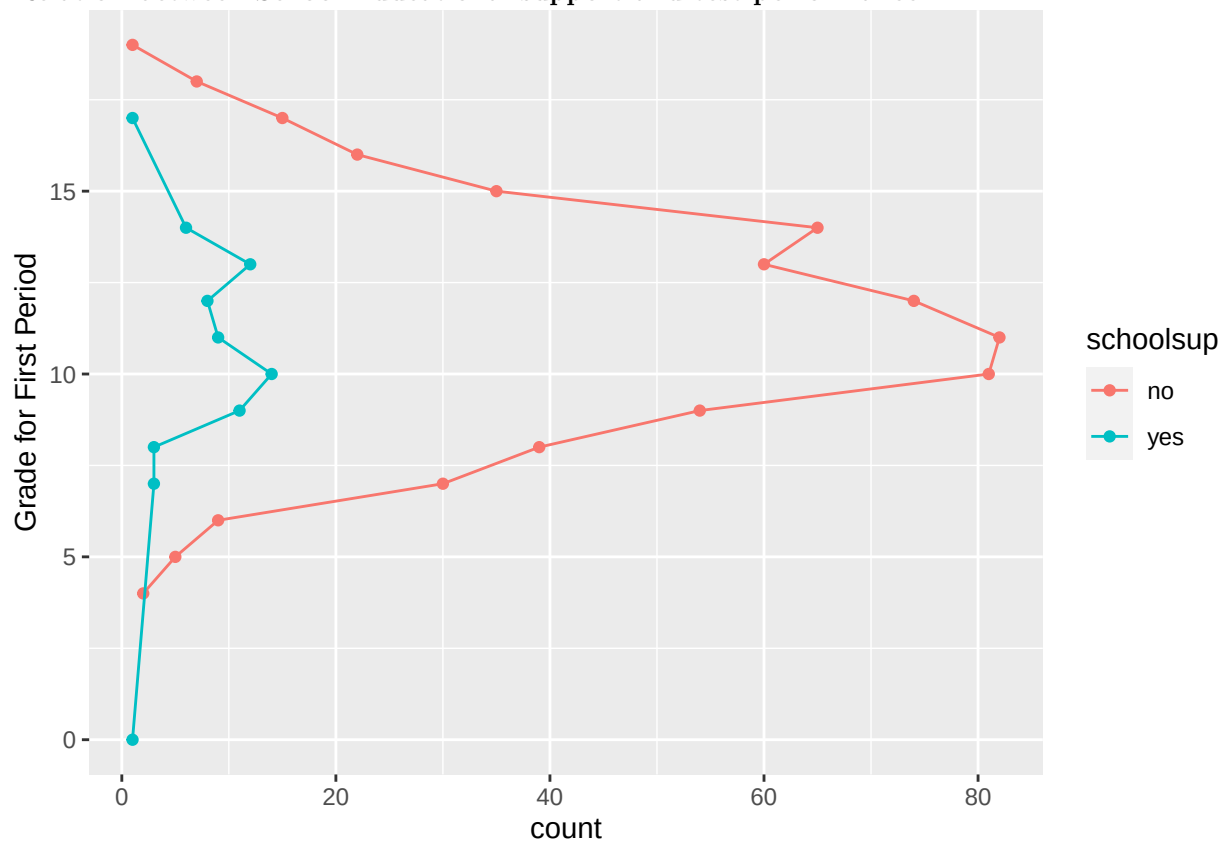


```
ggplot(data = student, mapping = aes(y=G3, colour=famsup)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Final Grade")
```

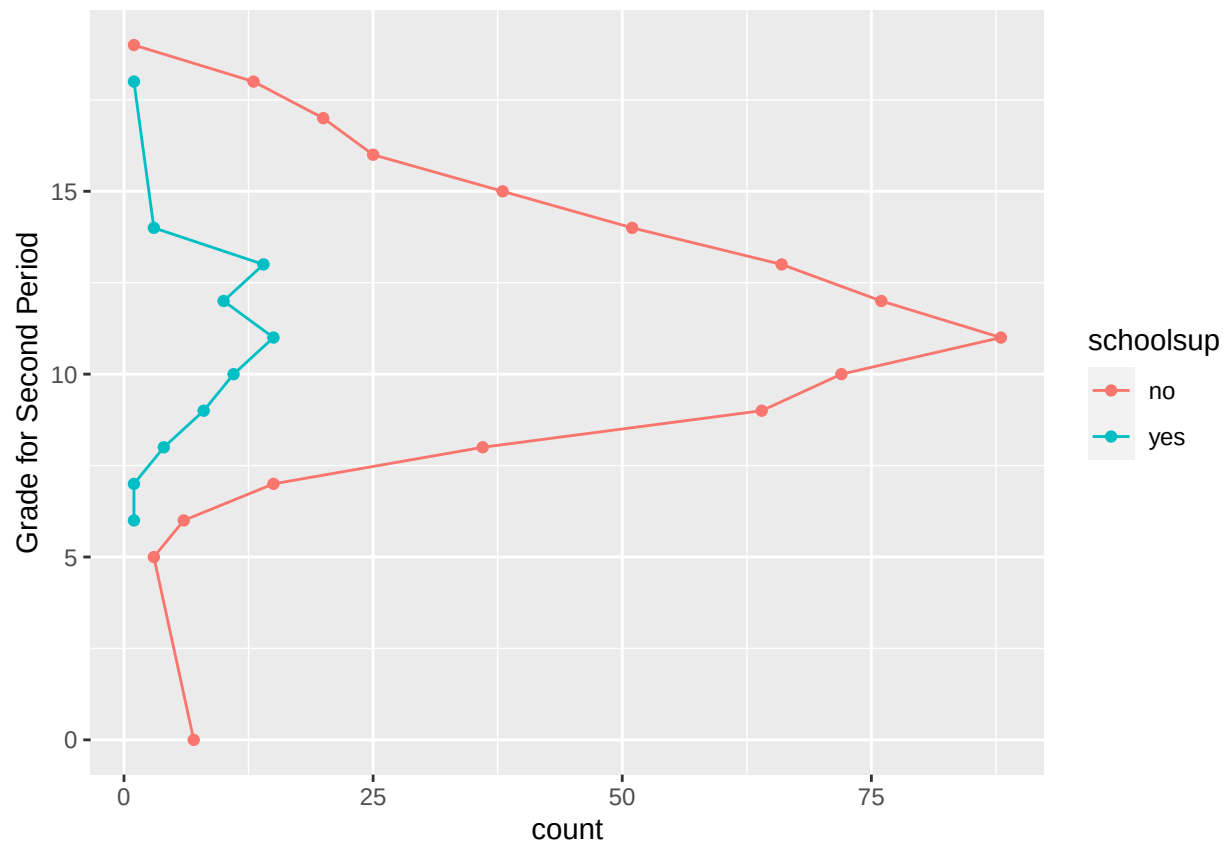


```
ggplot(data = student, mapping = aes(y=G1, colour=schoolsup)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for First Period")
```

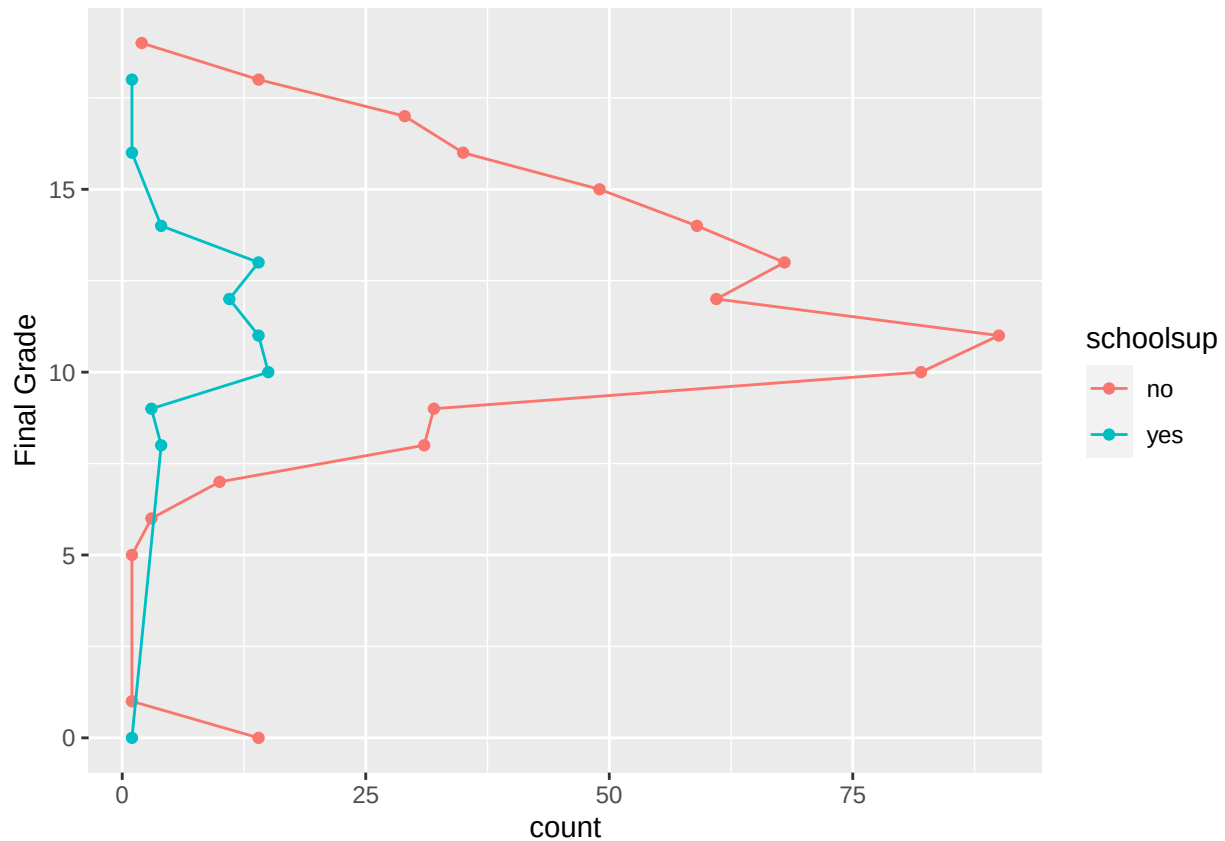
4) Relation between School Educational support and test performance



```
ggplot(data = student, mapping = aes(y=G2, colour=schoolsup)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for Second Period")
```



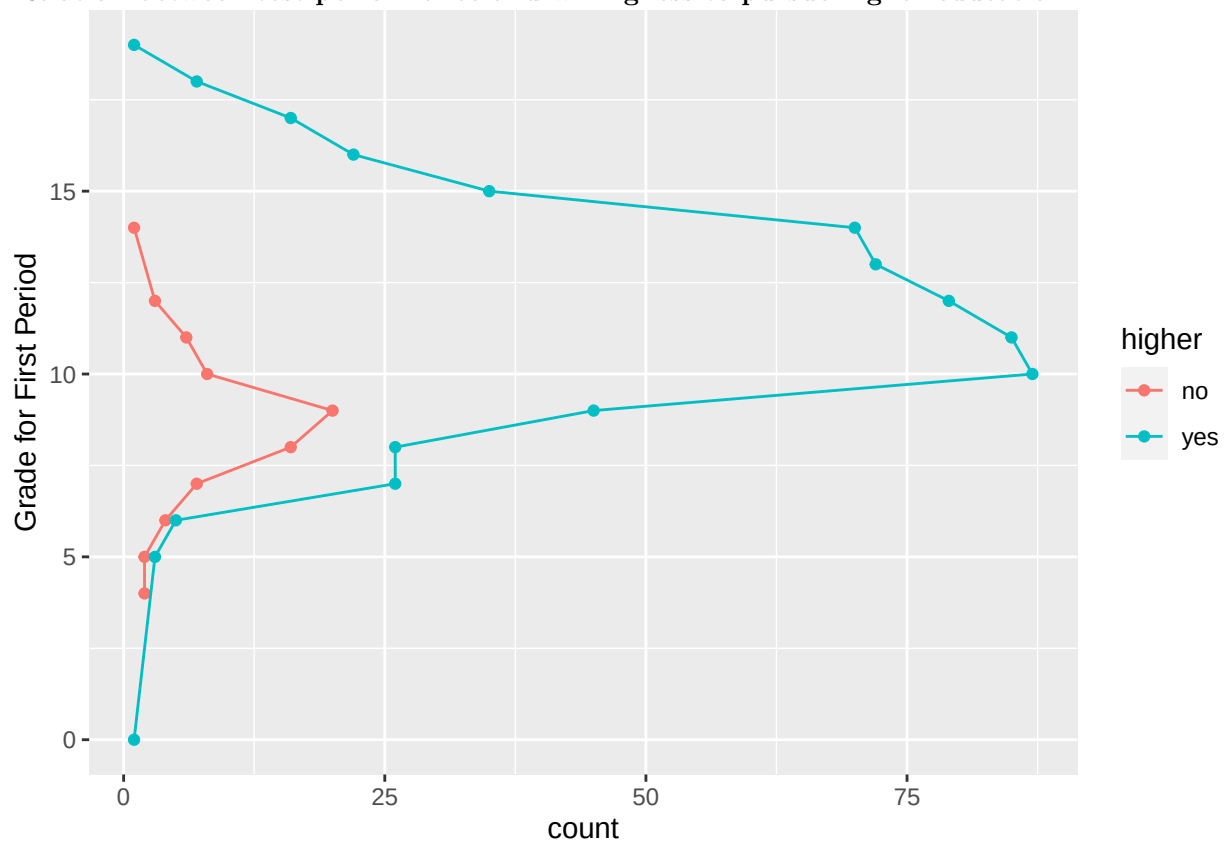
```
ggplot(data = student, mapping = aes(y=G3, colour=schoolsup)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Final Grade")
```



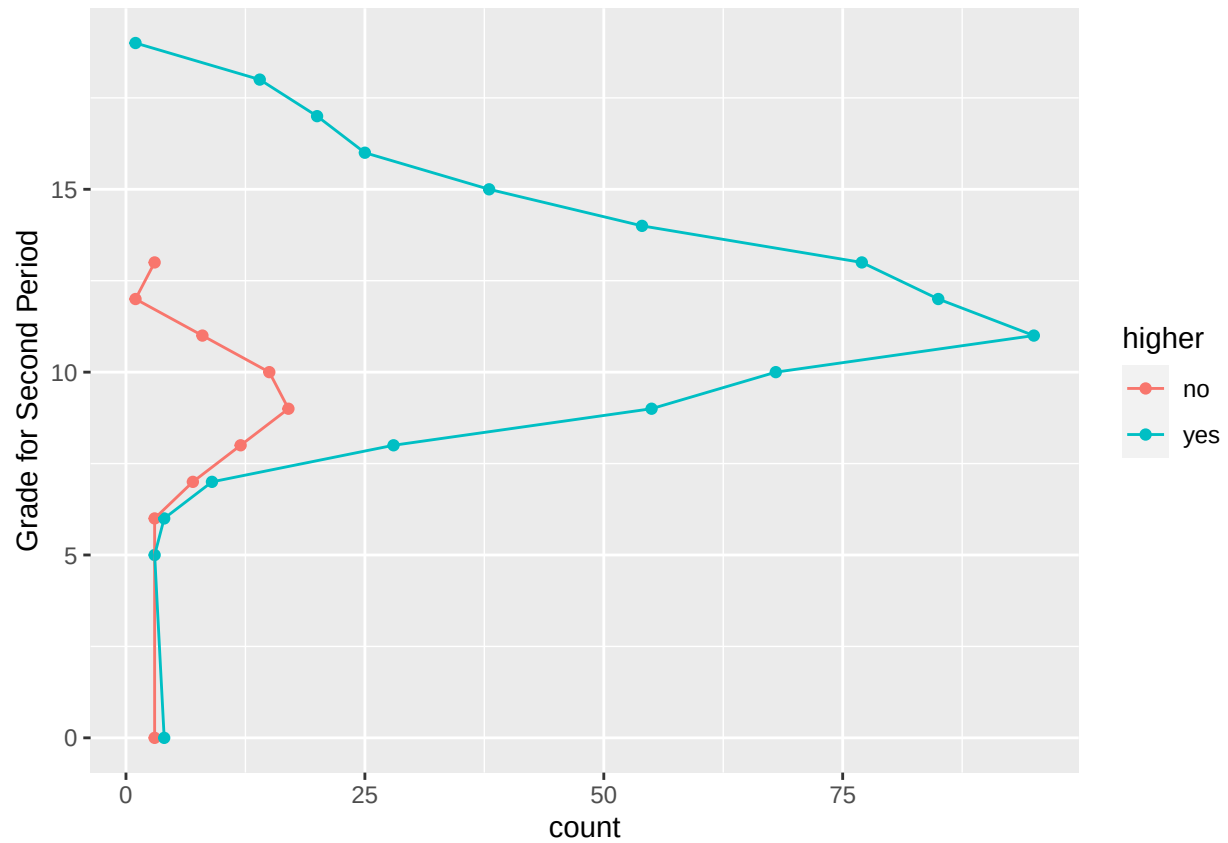
From the above 2 visualizations, we can say that students with family support for education are performing better and the students with school education support are performing better overall.

```
ggplot(data = student, mapping = aes(y=G1, colour=higher)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for First Period")
```

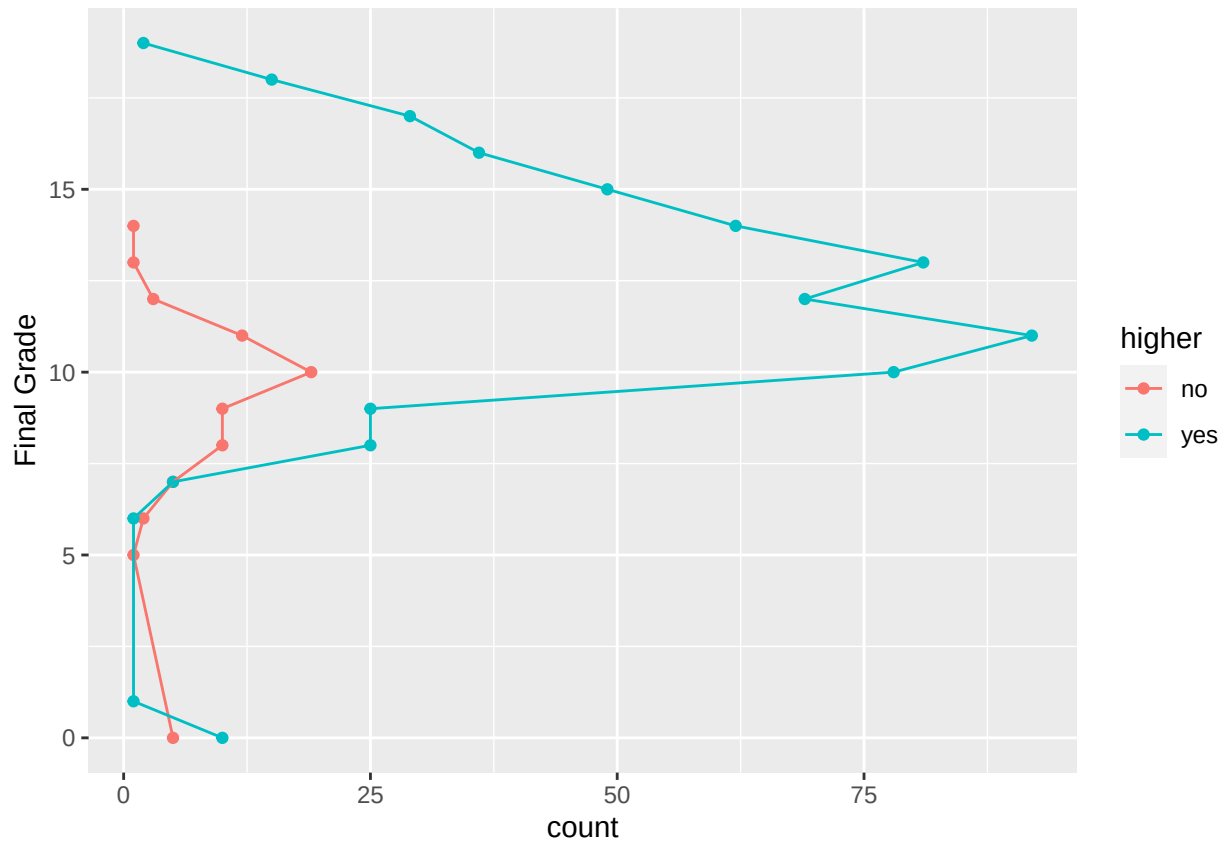
5) Relation between test performance and willingness to pursue higher education



```
ggplot(data = student, mapping = aes(y=G2, colour=higher)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for Second Period")
```



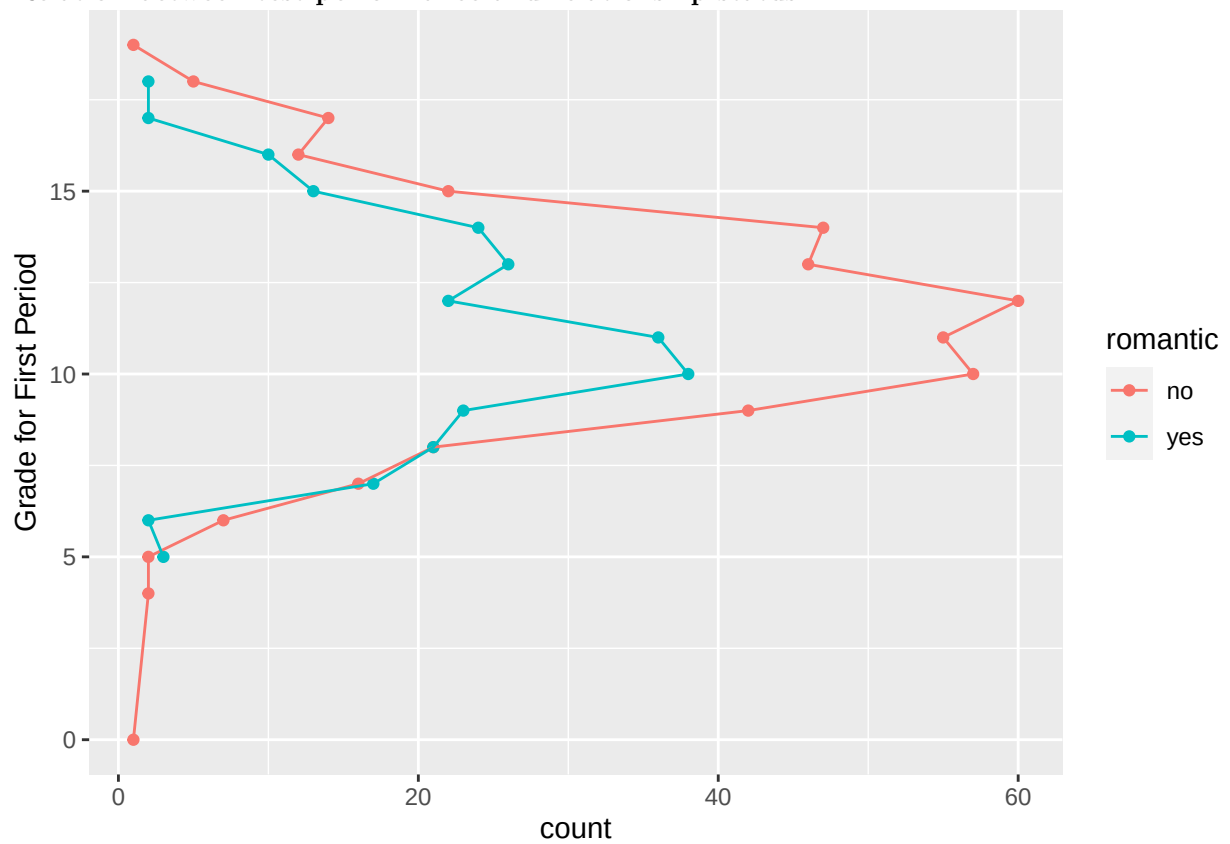
```
ggplot(data = student, mapping = aes(y=G3, colour=higher)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Final Grade")
```



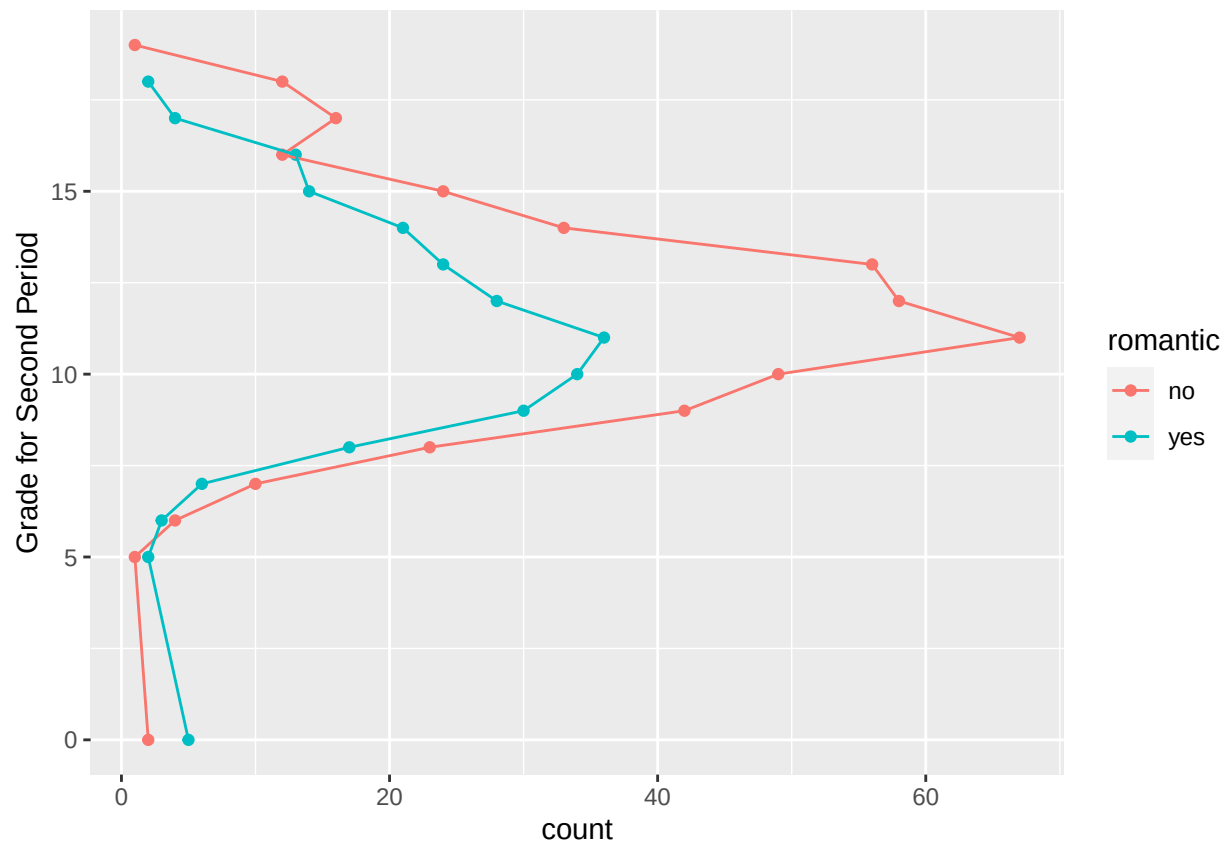
From the visualization, we can say that the students willing to pursue higher education are performing better overall.

```
ggplot(data = student, mapping = aes(y=G1, colour=romantic)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for First Period")
```

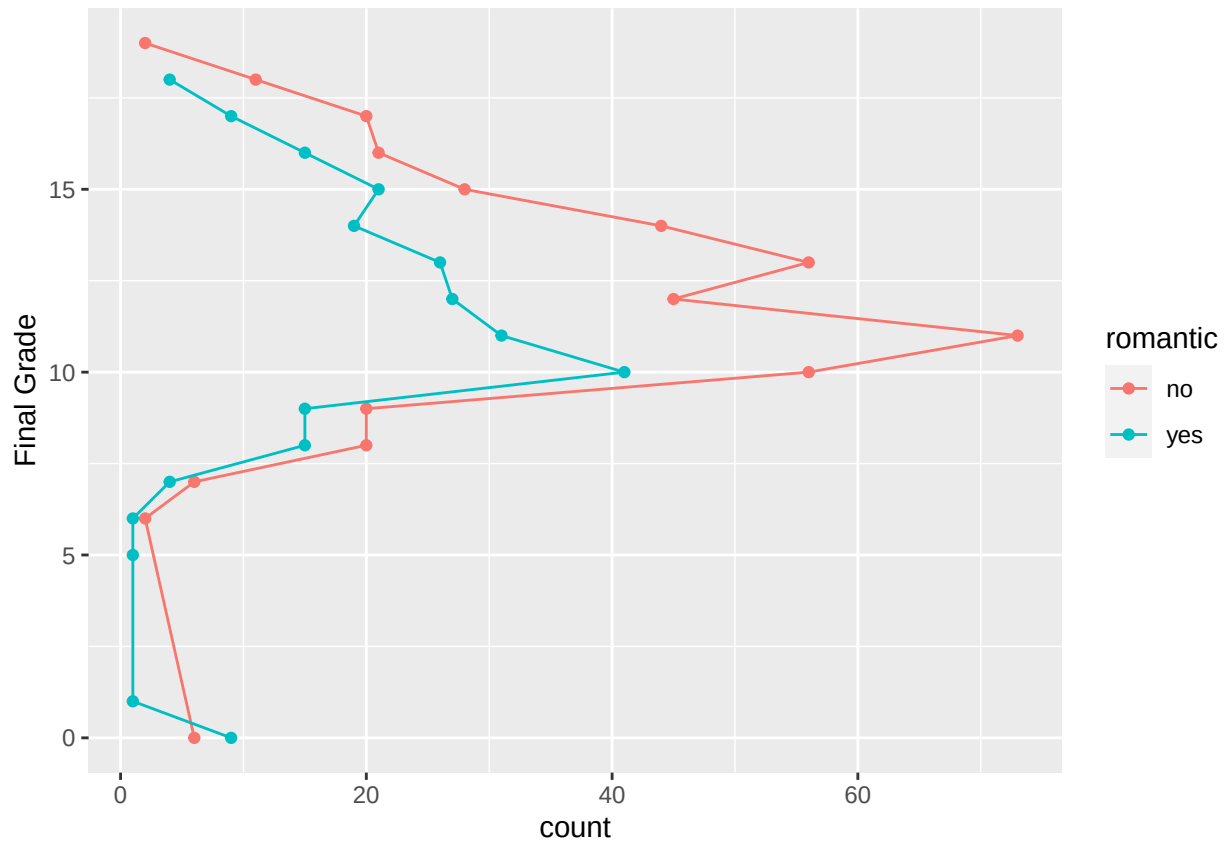
6) Relation between test performance and relationship status



```
ggplot(data = student, mapping = aes(y=G2, colour=romantic)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Grade for Second Period")
```



```
ggplot(data = student, mapping = aes(y=G3, colour=romantic)) +
  geom_point(stat = "count") +
  geom_line(stat = "count") +
  ylab("Final Grade")
```



From the visualization, we can say that majority of students who are not in a romantic relationship are performing better.

Data Analysis

PCA

PCA Analysis cannot be performed for this dataset because majority of the variables are categorical and not continuous in this dataset.

Hypothesis Testing

1) Is there a significant difference between final grades of students from different family sizes?

H0: There is no significant difference between final grades of students from different family sizes

H1: There is a significant difference between final grades of students from different family sizes

```
t.test(G3 ~ famsize, data = student, var.equal = TRUE)
```

```
##
## Two Sample t-test
##
## data: G3 by famsize
## t = -1.1462, df = 647, p-value = 0.2521
## alternative hypothesis: true difference in means between group GT3 and group LE3 is not equal to 0
```

```
## 95 percent confidence interval:
## -0.8638492 0.2270649
## sample estimates:
## mean in group GT3 mean in group LE3
##      11.81182      12.13021
```

As $p\text{-value} > 0.05$, we accept the null hypothesis. This means there is no significant difference between final grades of students from different family sizes.

2) Is there a significant difference between final grades of male and female students?

H0: There is no significant difference between final grades of male and female students

H1: There is a significant difference between final grades of male and female students

```
t.test(G3 ~ sex, data = student, var.equal = TRUE)
```

```
##
## Two Sample t-test
##
## data: G3 by sex
## t = 3.3109, df = 647, p-value = 0.0009815
## alternative hypothesis: true difference in means between group F and group M is not equal to 0
## 95 percent confidence interval:
## 0.3447659 1.3497314
## sample estimates:
## mean in group F mean in group M
##      12.25326      11.40602
```

As $p\text{-value} < 0.05$, we reject the null hypothesis. This means there is a significant difference between final grades of male and female students.

3) Is there a significant difference between final grades of students with extra educational support from school and those who don't?

H0: There is no significant difference between final grades of students with extra educational support from school and those who don't

H1: There is a significant difference between final grades of students with extra educational support from school and those who don't

```
t.test(G3 ~ schoolsup, data = student, var.equal = TRUE)
```

```
##
## Two Sample t-test
##
## data: G3 by schoolsup
## t = 1.6928, df = 647, p-value = 0.09097
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
## -0.1119752 1.5118436
## sample estimates:
## mean in group no mean in group yes
##      11.97935      11.27941
```

As $p\text{-value} > 0.05$, we accept the null hypothesis. This means there is no significant difference between final grades of students with extra educational support from school and those who don't.

4) Is there a significant difference between final grades of students with internet access and those who don't?

H0: There is no significant difference between final grades of students with internet access and those who don't

H1: There is a significant difference between final grades of students with internet access and those who don't

```
t.test(G3 ~ internet, data = student, var.equal = TRUE)
```

```
##
## Two Sample t-test
##
## data: G3 by internet
## t = -3.8597, df = 647, p-value = 0.0001249
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
## -1.7293279 -0.5630735
## sample estimates:
## mean in group no mean in group yes
## 11.02649 12.17269
```

As $p\text{-value} < 0.05$, we reject the null hypothesis. This means there is a significant difference between final grades of students with internet access and those who don't.

5) Is there a significant difference between final grades of students in a romantic relationship and those who aren't?

H0: There is no significant difference between final grades of students in a romantic relationship and those who aren't

H1: There is a significant difference between final grades of students in a romantic relationship and those who aren't

```
t.test(G3 ~ romantic, data = student, var.equal = TRUE)
```

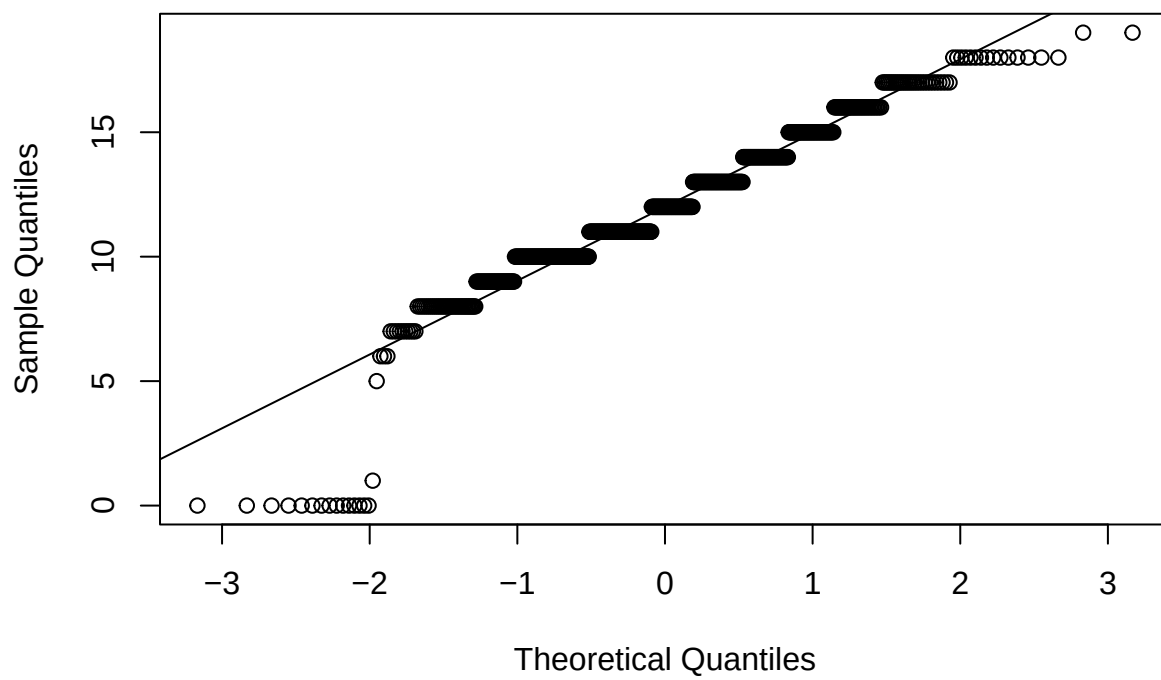
```
##
## Two Sample t-test
##
## data: G3 by romantic
## t = 2.3136, df = 647, p-value = 0.021
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
## 0.09170316 1.12080832
## sample estimates:
## mean in group no mean in group yes
## 12.12927 11.52301
```

As $p\text{-value} < 0.05$, we reject the null hypothesis. This means there is a significant difference between final grades of students in a romantic relationship and those who aren't.

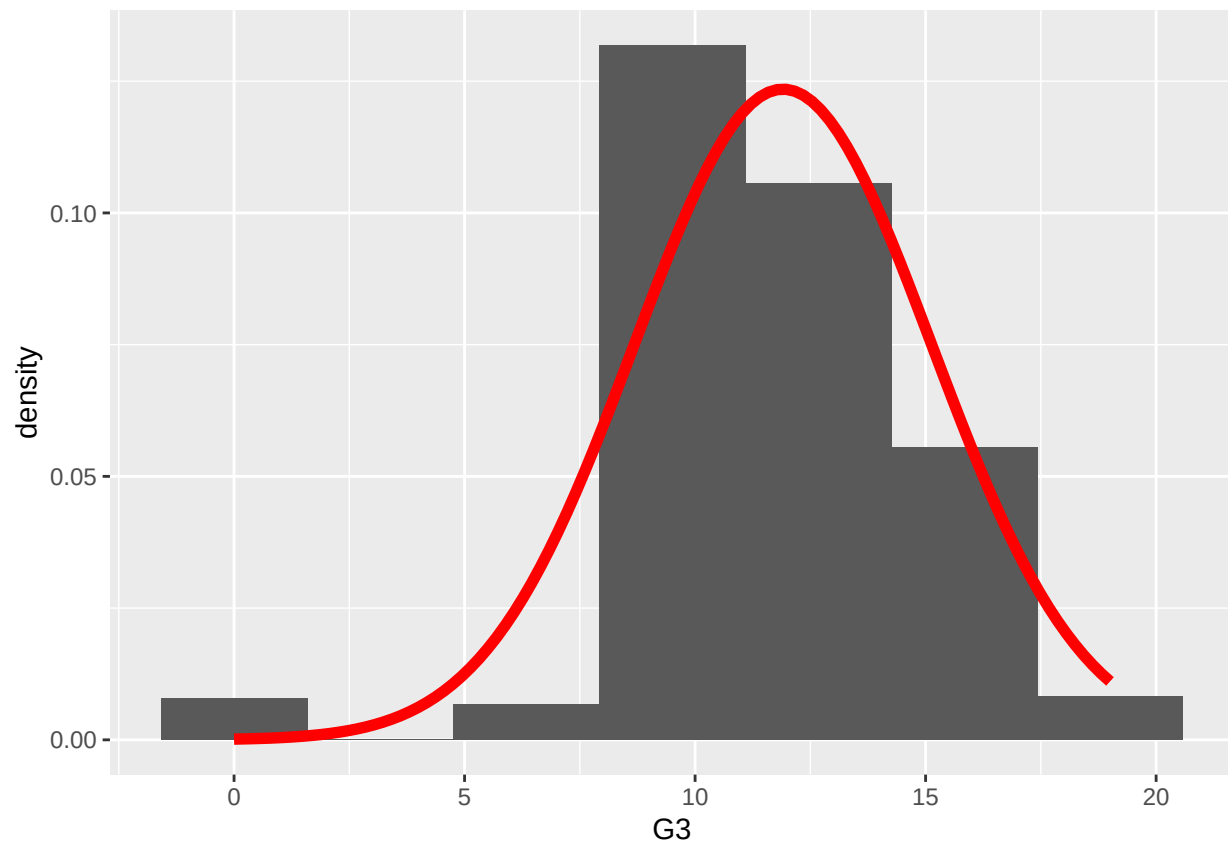
```
qqnorm(student$G3)
qqline(student$G3, distribution = qnorm )
```

Linear Regression

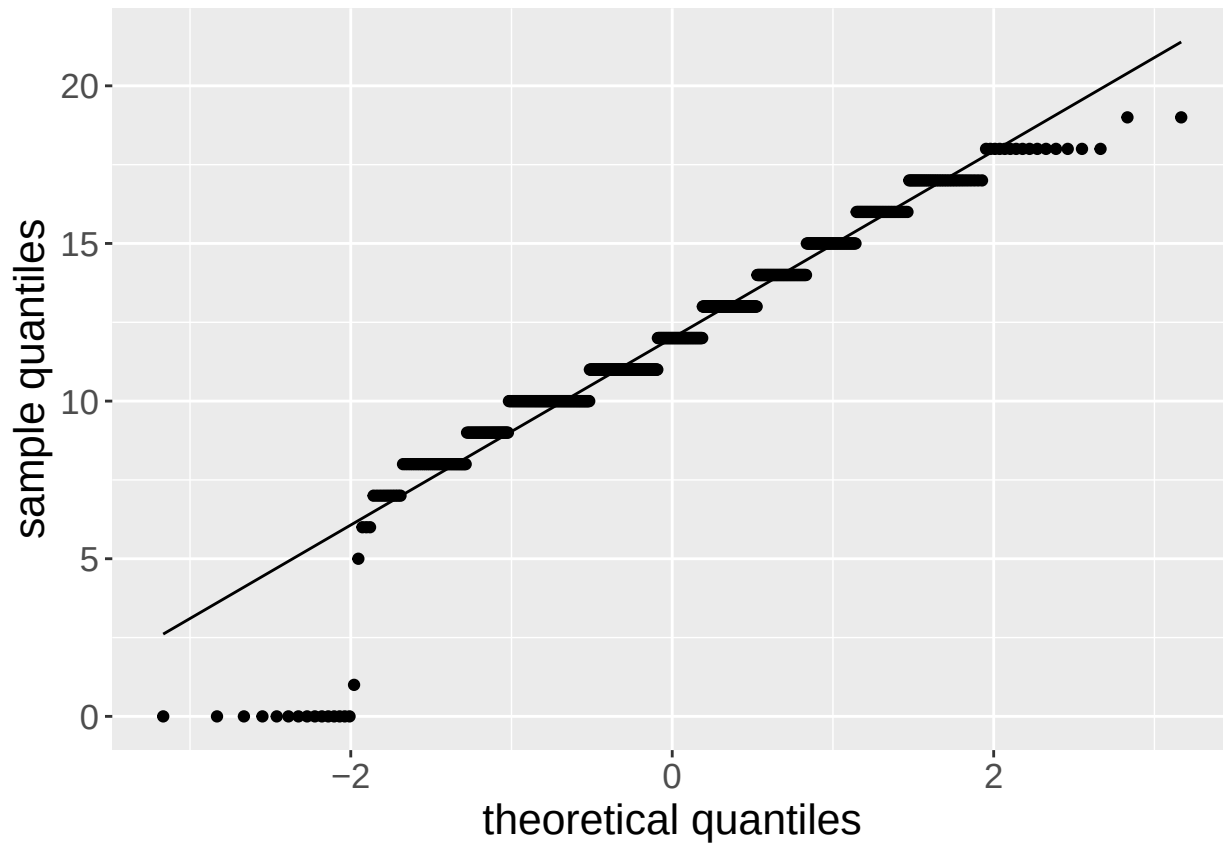
Normal Q-Q Plot



```
ggplot(data = student, aes(G3)) + geom_histogram(mapping = aes(x = G3, y = stat(density)), bins = 7) +
  stat_function(
    fun = dnorm,
    args = list(mean = mean(student$G3), sd = sd(student$G3)),
    lwd = 2,
    col = 'red')
```



```
wind <- ggplot(data = student, aes(sample = G3))  
wind + stat_qq(distribution = stats::qnorm) + stat_qq_line() +  
  labs(y = 'sample quantiles', x = 'theoretical quantiles') +  
  theme(text = element_text(size = 16))
```



```
linear = lm(G3~studytime+schoolsup+sex+school+higher, data=student)
print(linear)
```

```
##
## Call:
## lm(formula = G3 ~ studytime + schoolsup + sex + school + higher,
##     data = student)
##
## Coefficients:
## (Intercept)      studytime  schoolsupyes          sexM      schoolMS
##      9.2791         0.5905        -1.5602        -0.7838        -1.7223
##      higheryes
##      2.8771
```

```
summary(linear)
```

```
##
## Call:
## lm(formula = G3 ~ studytime + schoolsup + sex + school + higher,
##     data = student)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
##	-11.615	-1.615	0.037	1.759	8.169

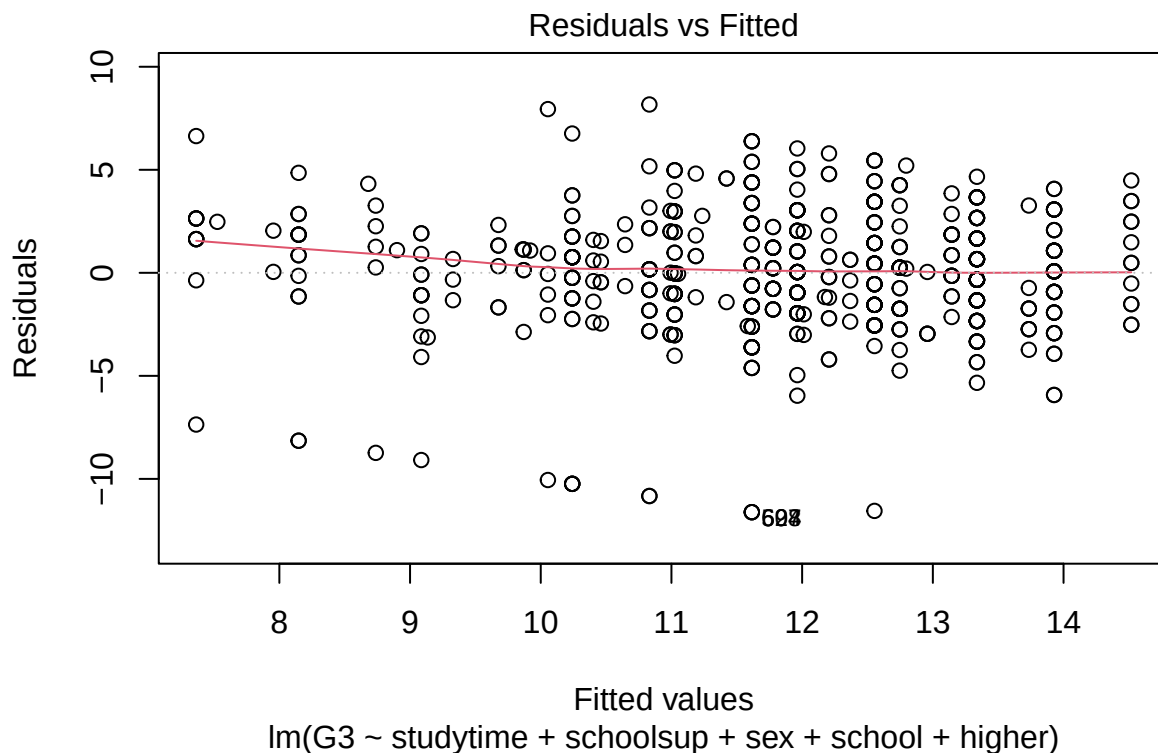
```
##
```

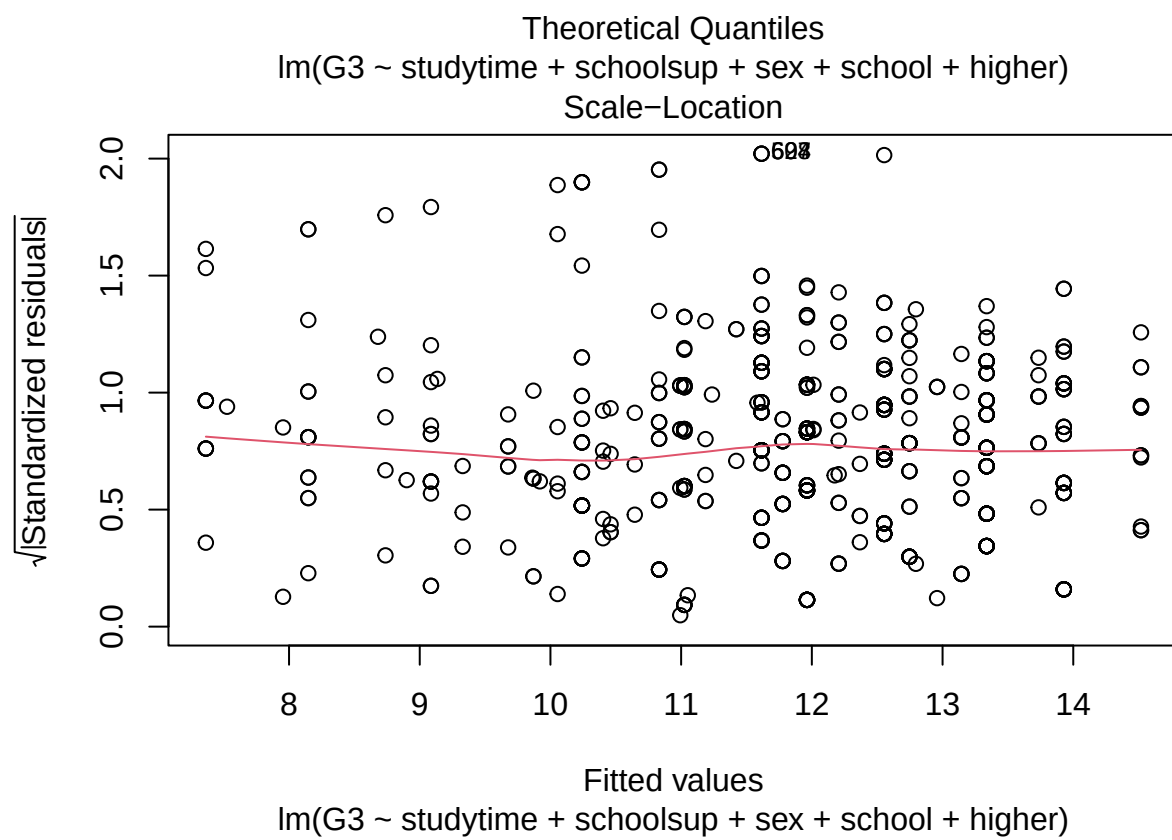
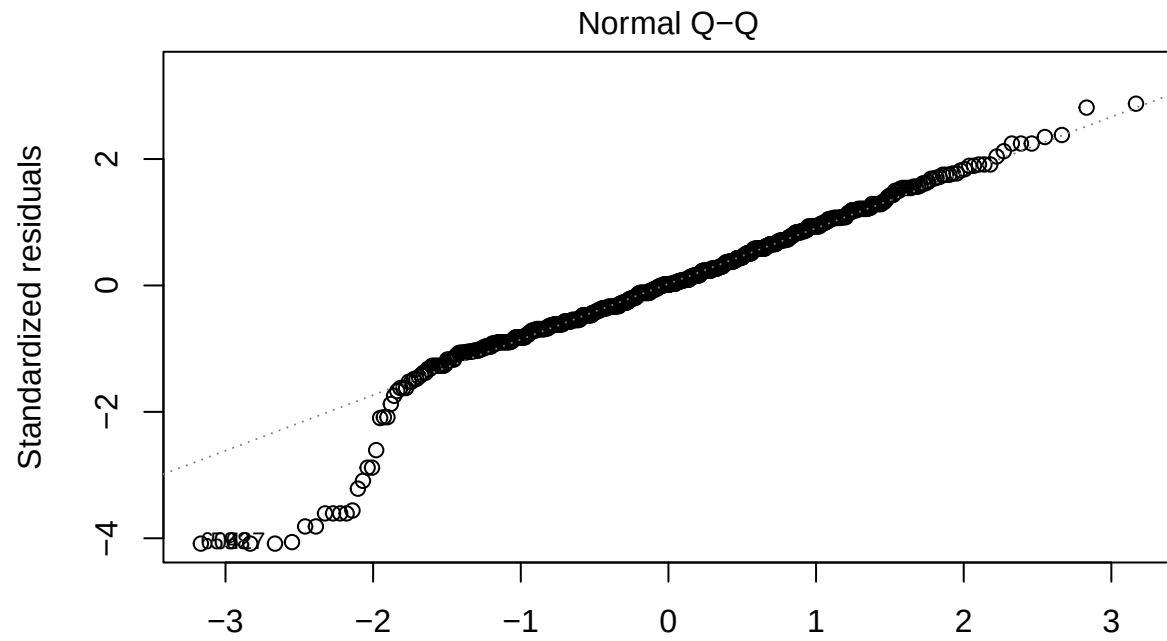
```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.2791     0.4624  20.067 < 2e-16 ***
## studytime     0.5905     0.1418   4.165 3.54e-05 ***
## schoolsupyes  -1.5602     0.3722  -4.192 3.15e-05 ***
## sexM          -0.7838     0.2357  -3.325 0.000933 ***
## schoolMS      -1.7223     0.2422  -7.112 3.06e-12 ***
## higheryes     2.8771     0.3730   7.714 4.67e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.852 on 643 degrees of freedom
## Multiple R-squared:  0.2266, Adjusted R-squared:  0.2206
## F-statistic: 37.69 on 5 and 643 DF,  p-value: < 2.2e-16
```

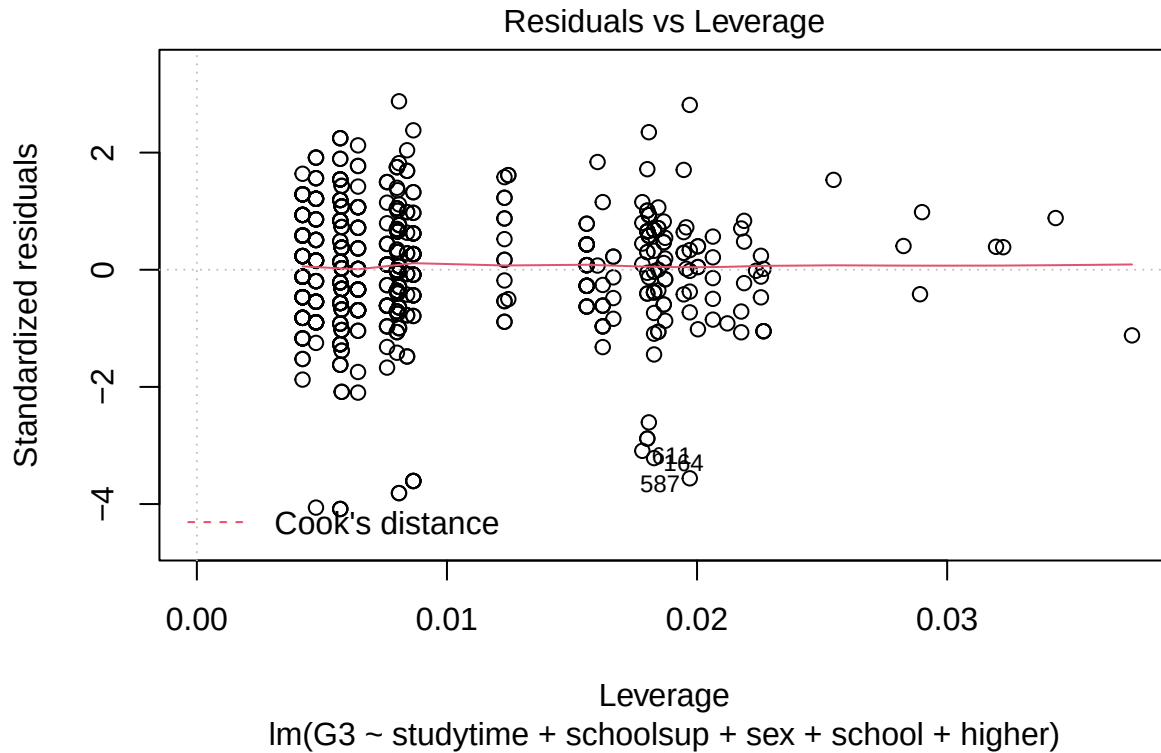
We can consider a linear model to be statistically significant only when the p-Values are less than the pre-determined statistical significance level of 0.05. This is true from the summary shown above. The more the stars beside the variable's p-Value, the more significant the variable.

The Adjusted R-squared value is 0.2206 which means that the accuracy of predicting the final grade if the student from the given parameters is 22.06%. This means it is not that accurate of a predictive model.

```
plot(linear)
```







The first plot (residuals vs. fitted values) is a simple scatterplot between residuals and predicted values. It should look more or less random. This is more or less what we see above, with the exception of a few outliers.

The second plot (normal Q-Q) is a normal probability plot. It gives a straight line if the errors are distributed normally, but a few points deviate from the straight line.

The third plot (Scale-Location), like the first, should look random. No patterns. The one above looks random more or less.

The last plot (Cook's distance) tells us which points have the greatest influence on the regression (leverage points). We see a few points that have great influence on the model.

Conclusions

From the initial visualizations, I was able to come to the following conclusions:

- 1) The course curriculum provided by the school was the major reason for switching schools.
- 2) The students studying for longer period of time are getting better grades overall.
- 3) The students that have an extra financial help from the school and/or family are getting better grades overall.
- 4) The students who are intending to pursue higher education are getting better grades overall.
- 5) The students who are in a romantic relations are performing poorly as compared to the other students.

The multiple hypothesis testing which I performed lead me to the following conclusions:

- 1) There is no significant difference between final grades of students from different family sizes.

- 2) There is a significant difference between final grades of male and female students.
- 3) There is no significant difference between final grades of students with extra educational support from school and those who don't.
- 4) There is a significant difference between final grades of students with internet access and those who don't.
- 5) There is a significant difference between final grades of students in a romantic relationship and those who aren't.

The linear regression was performed on the data using the predictors study time, intention to pursue higher education, sex, extra financial support from school and current school to predict the final grade of the student. This showed that the all predictors used were significant in the prediction of the final grade. Unfortunately, the adjusted R-squared value of the fit was very low for a linear regression model to make accurate predictions. Hence, this model cannot be used to make the prediction for the final grade of the student.

Data analysis that can be performed on this dataset is very limited because the majority of the variables being categorical. For future research, the values for all the categorical variables can be converted to continuous to perform data analysis to get accurate results. The predictions can improve and become more accurate using continuous data.

References

- 1) <http://archive.ics.uci.edu/ml/datasets/Student+Performance>
- 2) <http://www3.dsi.uminho.pt/pcortez/student.pdf>

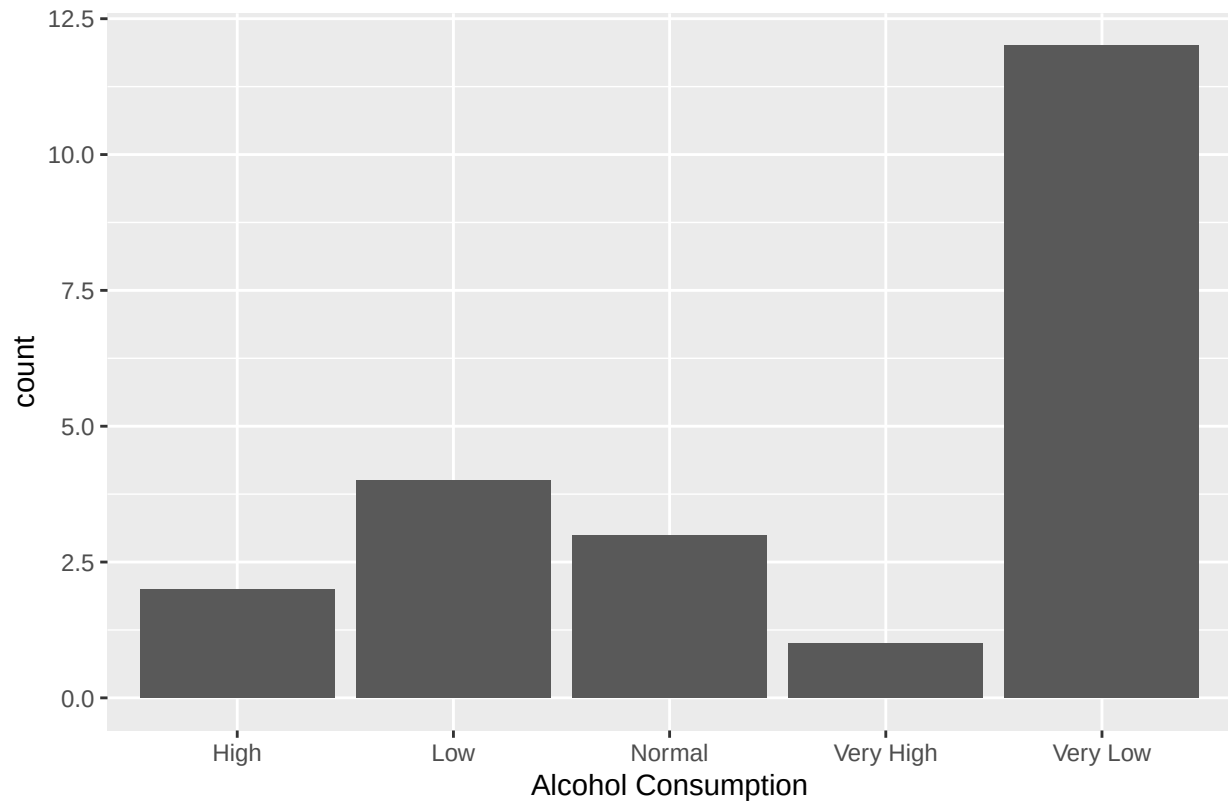
Appendix

Additional plots which are not relevant to the project.

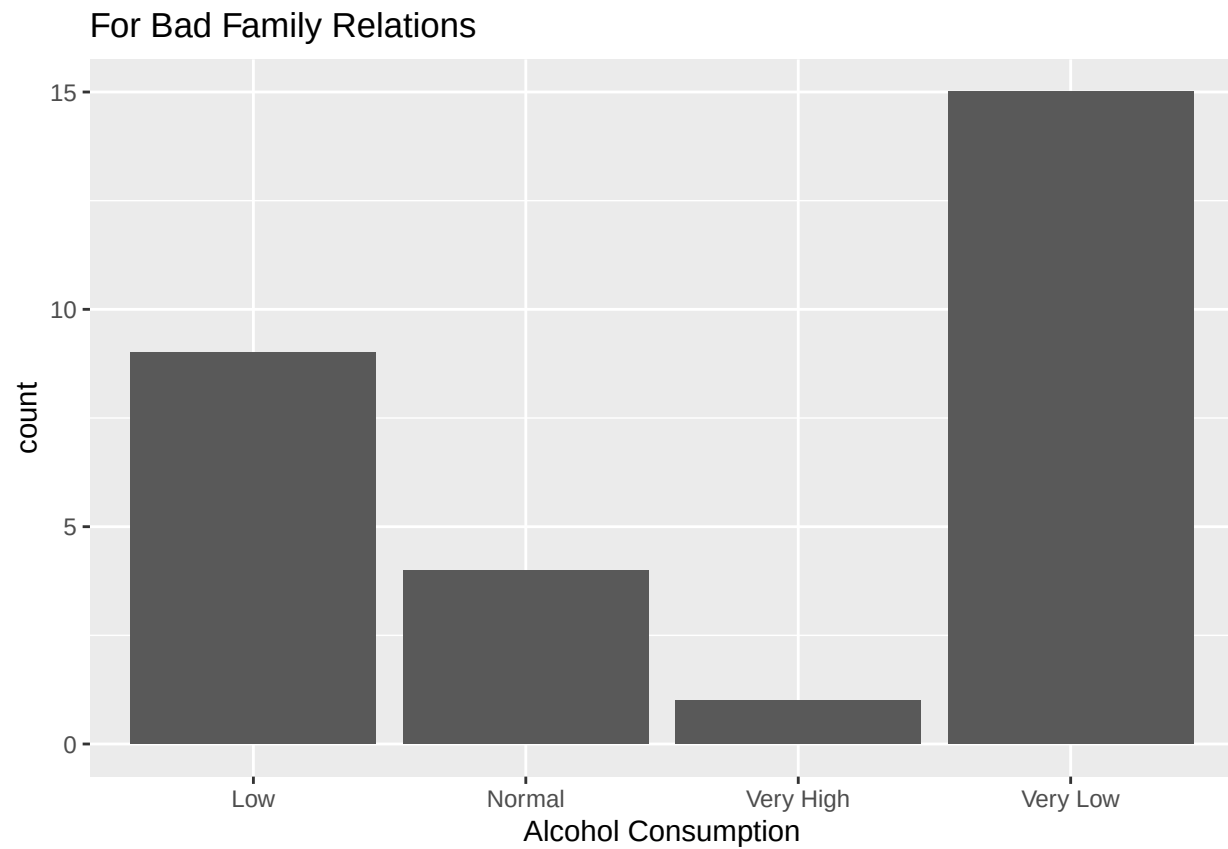
```
student.f1 <- student[student$famrel == "1",]
student.f2 <- student[student$famrel == "2",]
student.f3 <- student[student$famrel == "3",]
student.f4 <- student[student$famrel == "4",]
student.f5 <- student[student$famrel == "5",]

ggplot(data = student.f1, mapping = aes(x=Dalc.cat)) +
  ggtitle("For Very Bad Family Relations") +
  xlab("Alcohol Consumption") +
  geom_bar()
```

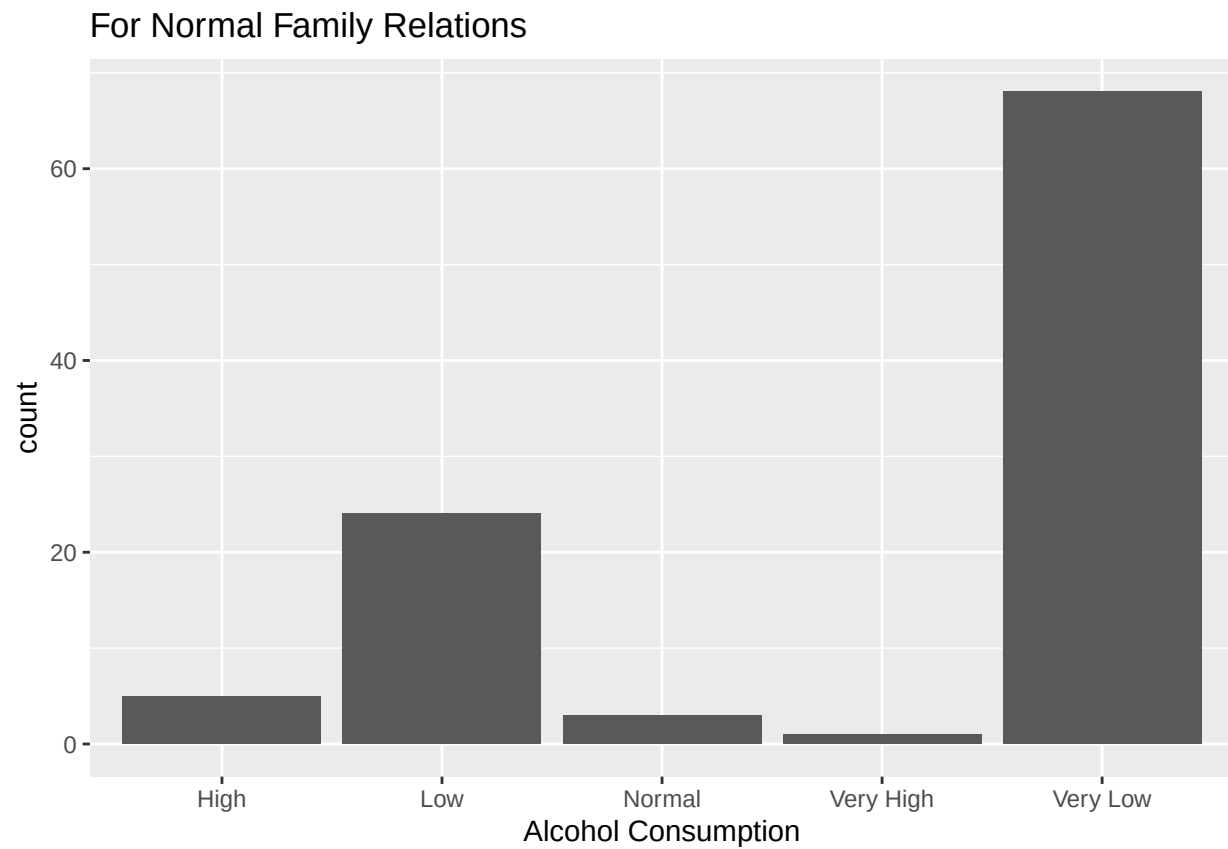
1) Relation between family relations and alcohol consumption
For Very Bad Family Relations



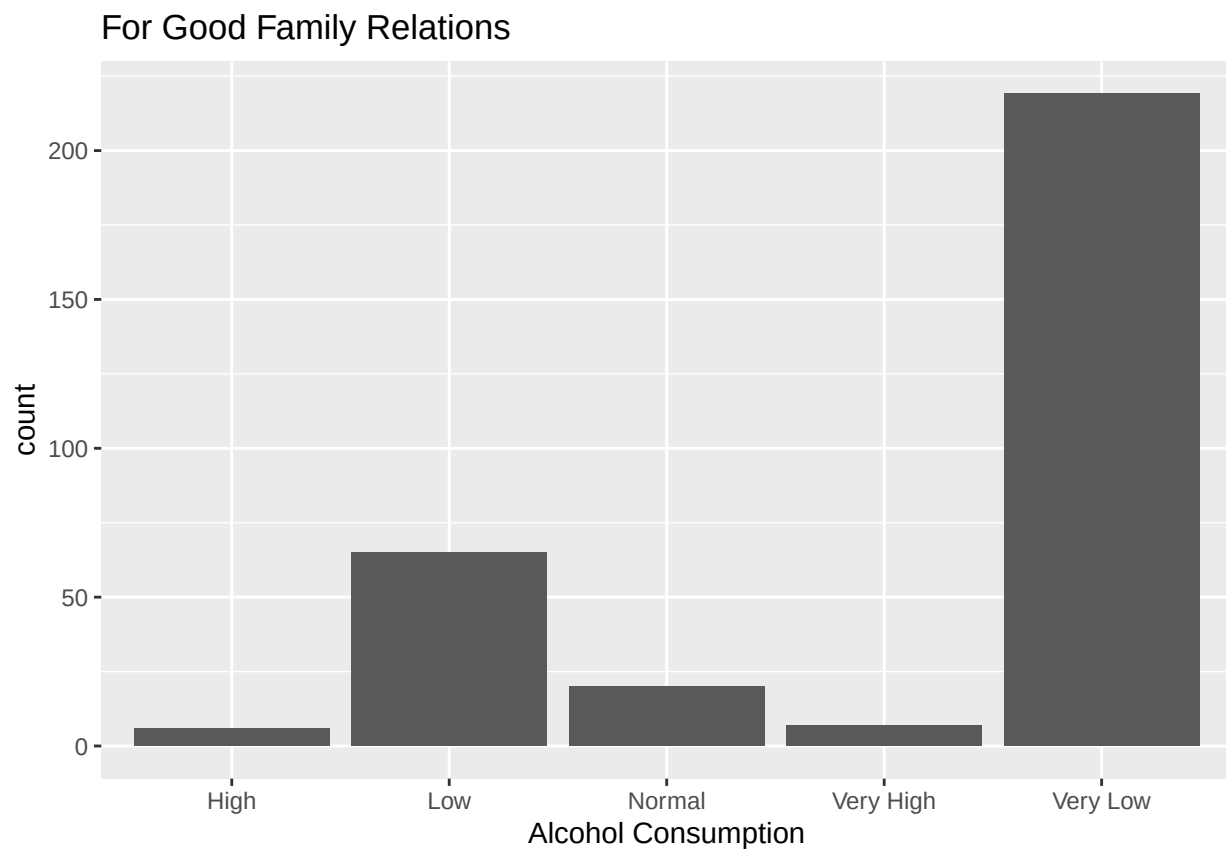
```
ggplot(data = student.f2, mapping = aes(x=Dalc.cat)) +  
  ggtitle("For Bad Family Relations") +  
  xlab("Alcohol Consumption") +  
  geom_bar()
```



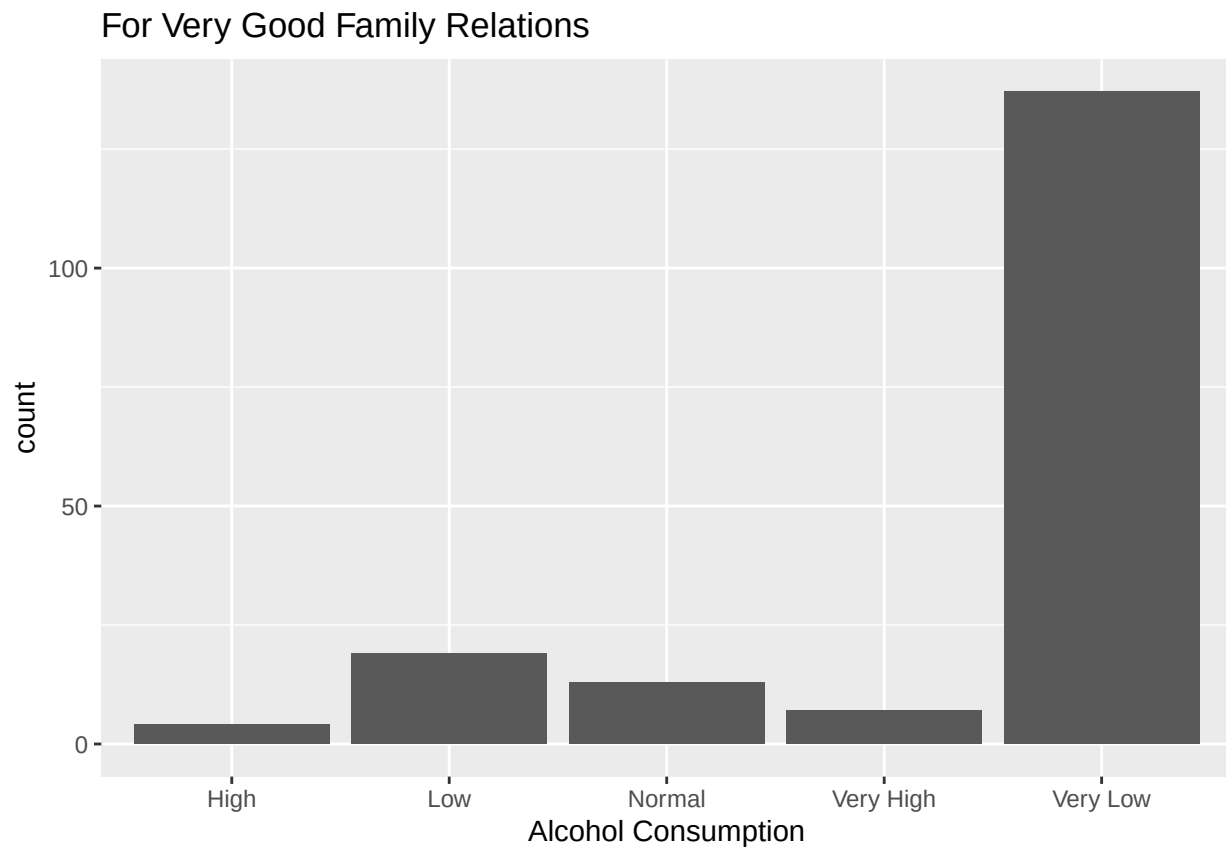
```
ggplot(data = student.f3, mapping = aes(x=Dalc.cat)) +  
  ggtitle("For Normal Family Relations") +  
  xlab("Alcohol Consumption") +  
  geom_bar()
```



```
ggplot(data = student.f4, mapping = aes(x=Dalc.cat)) +  
  ggtitle("For Good Family Relations") +  
  xlab("Alcohol Consumption") +  
  geom_bar()
```



```
ggplot(data = student.f5, mapping = aes(x=Dalc.cat)) +  
  ggtitle("For Very Good Family Relations") +  
  xlab("Alcohol Consumption") +  
  geom_bar()
```

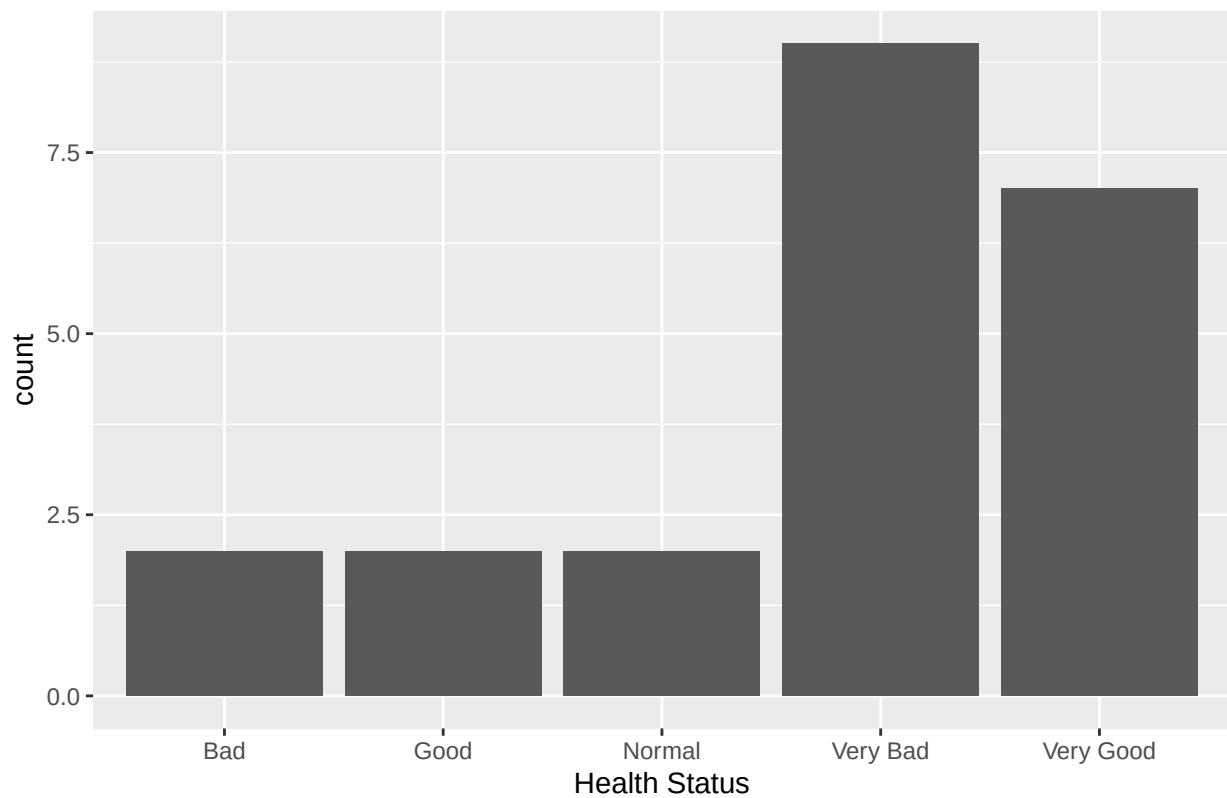


From the visualization, we cannot reach a standard conclusion. This visualization is inconclusive.

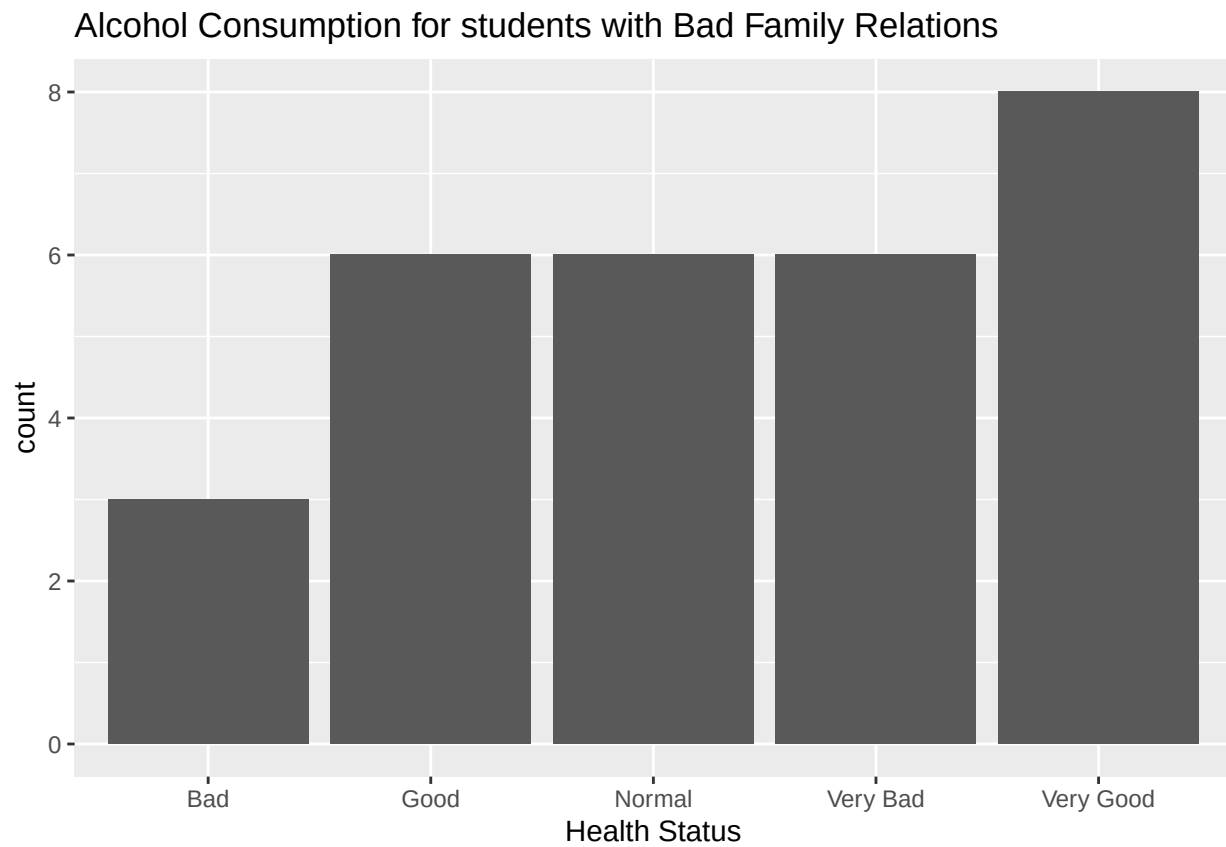
```
ggplot(data = student.f1, mapping = aes(x=health.cat)) +  
  ggtitle("Alcohol Consumption for students with Very Bad Family Relations") +  
  xlab("Health Status") +  
  geom_bar()
```

2) Relation between family relations and health status

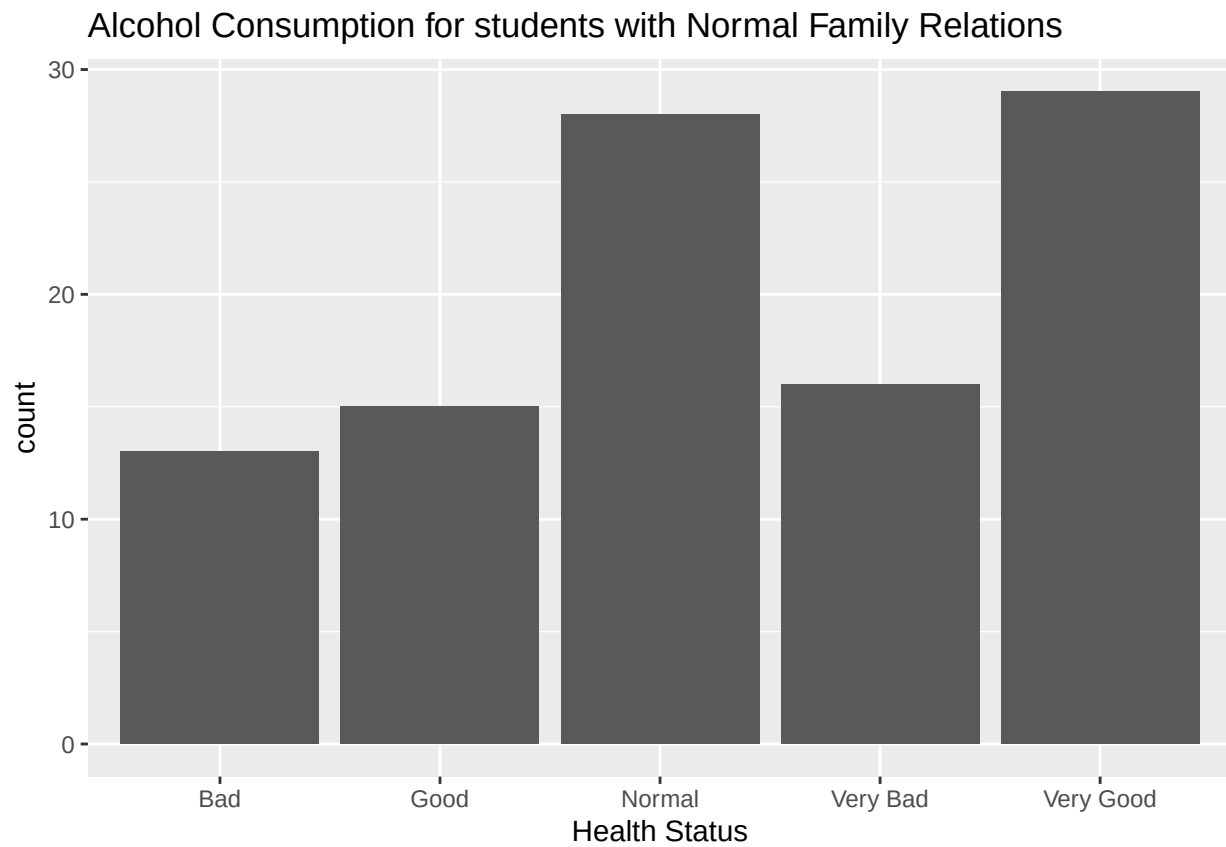
Alcohol Consumption for students with Very Bad Family Relations



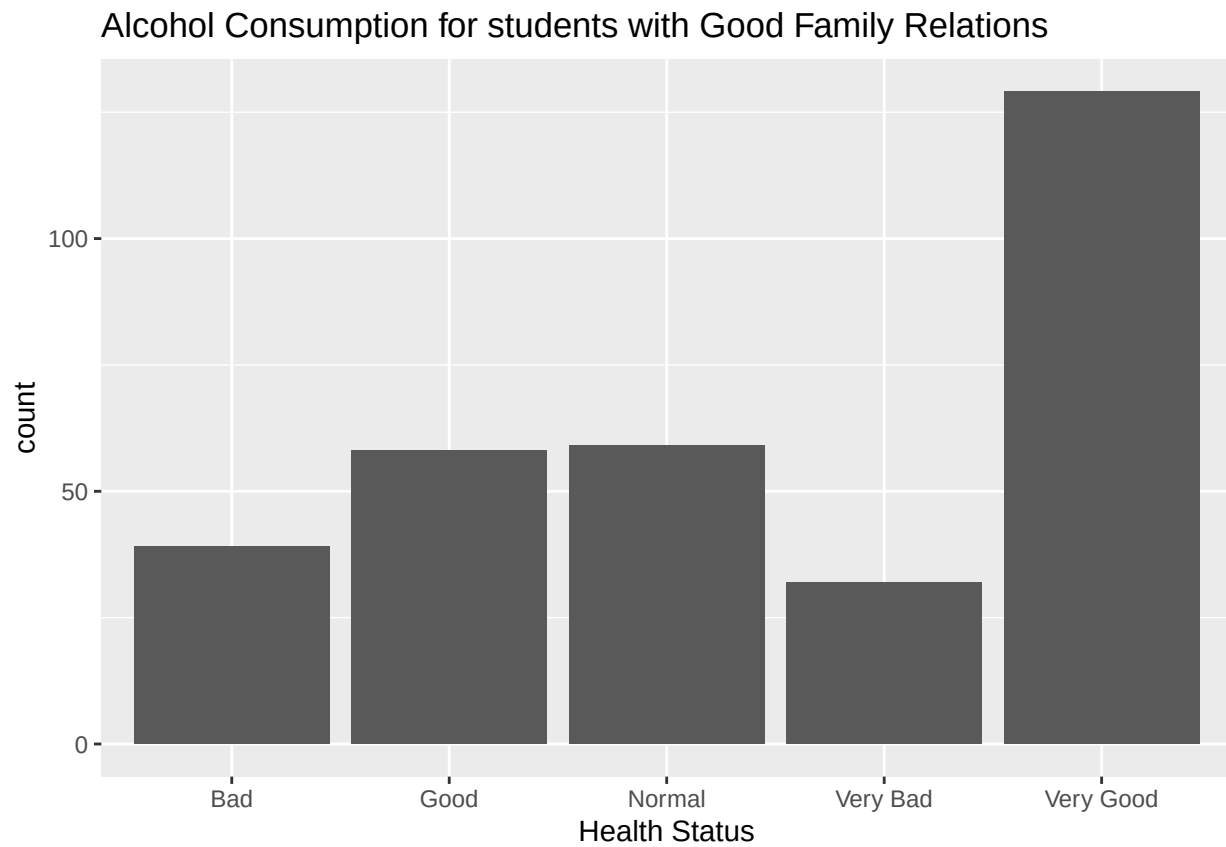
```
ggplot(data = student.f2, mapping = aes(x=health.cat)) +  
  ggtitle("Alcohol Consumption for students with Bad Family Relations") +  
  xlab("Health Status") +  
  geom_bar()
```



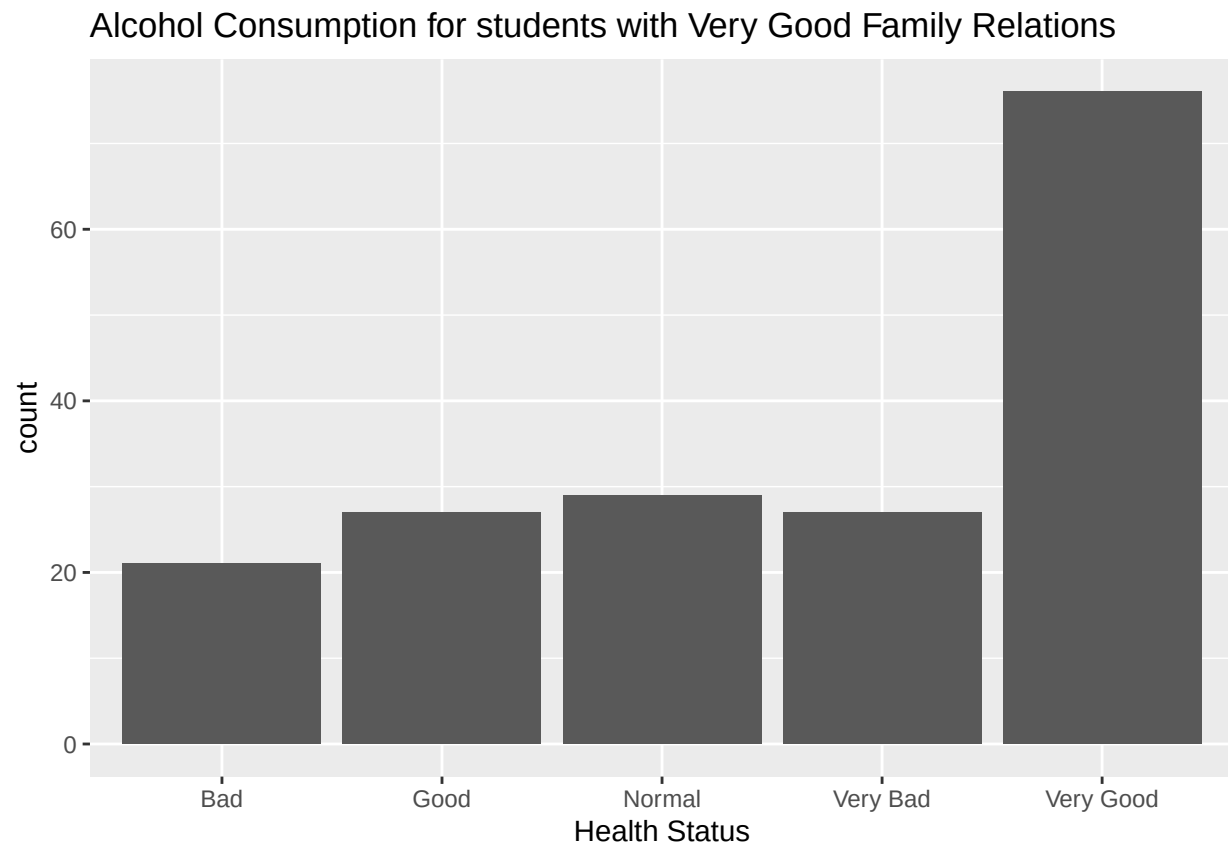
```
ggplot(data = student.f3, mapping = aes(x=health.cat)) +  
  ggtitle("Alcohol Consumption for students with Normal Family Relations") +  
  xlab("Health Status") +  
  geom_bar()
```



```
ggplot(data = student.f4, mapping = aes(x=health.cat)) +  
  ggtitle("Alcohol Consumption for students with Good Family Relations") +  
  xlab("Health Status") +  
  geom_bar()
```



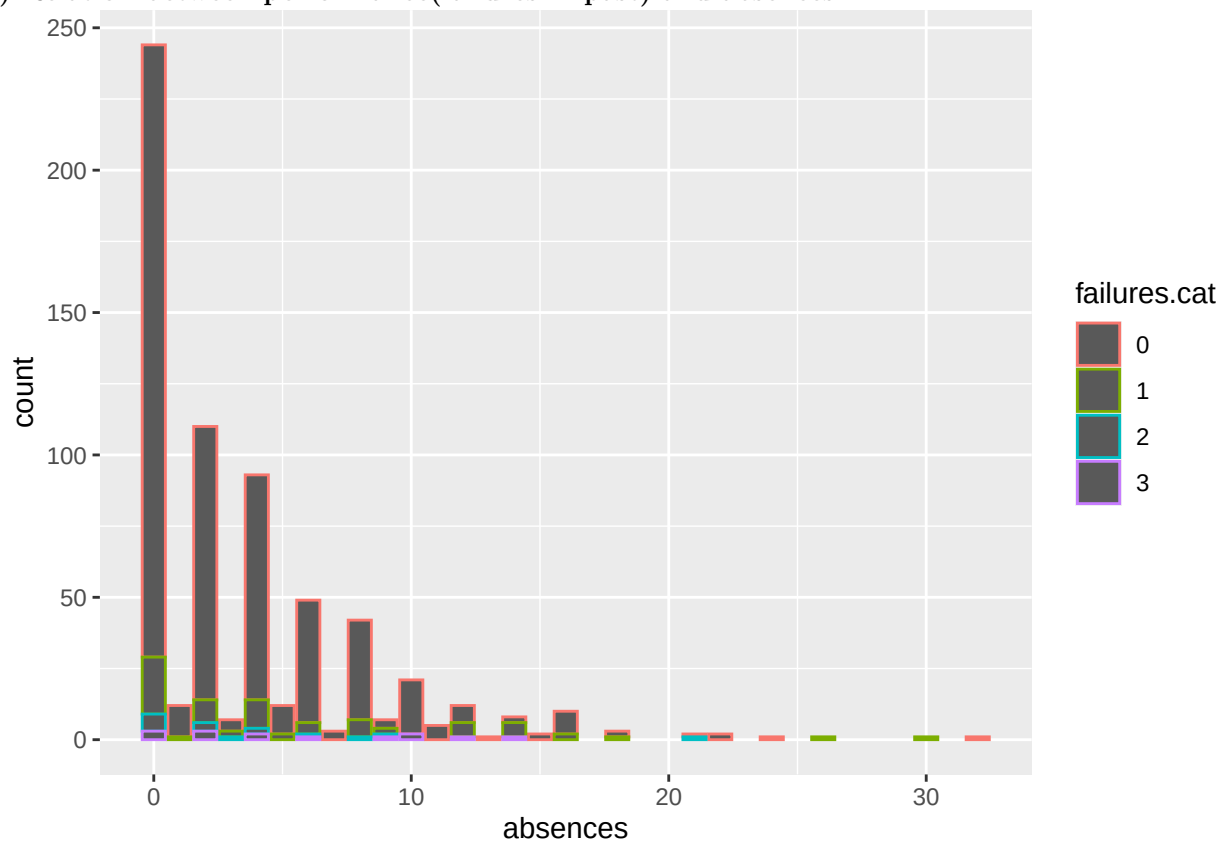
```
ggplot(data = student.f5, mapping = aes(x=health.cat)) +  
  ggtitle("Alcohol Consumption for students with Very Good Family Relations") +  
  xlab("Health Status") +  
  geom_bar()
```



From the visualization, we can say that very bad family relations have more bad health status whereas very good family relations have good health status.

```
ggplot(data = student, mapping = aes(x=absences, colour=failures.cat)) +  
  geom_bar(stat = "count")
```

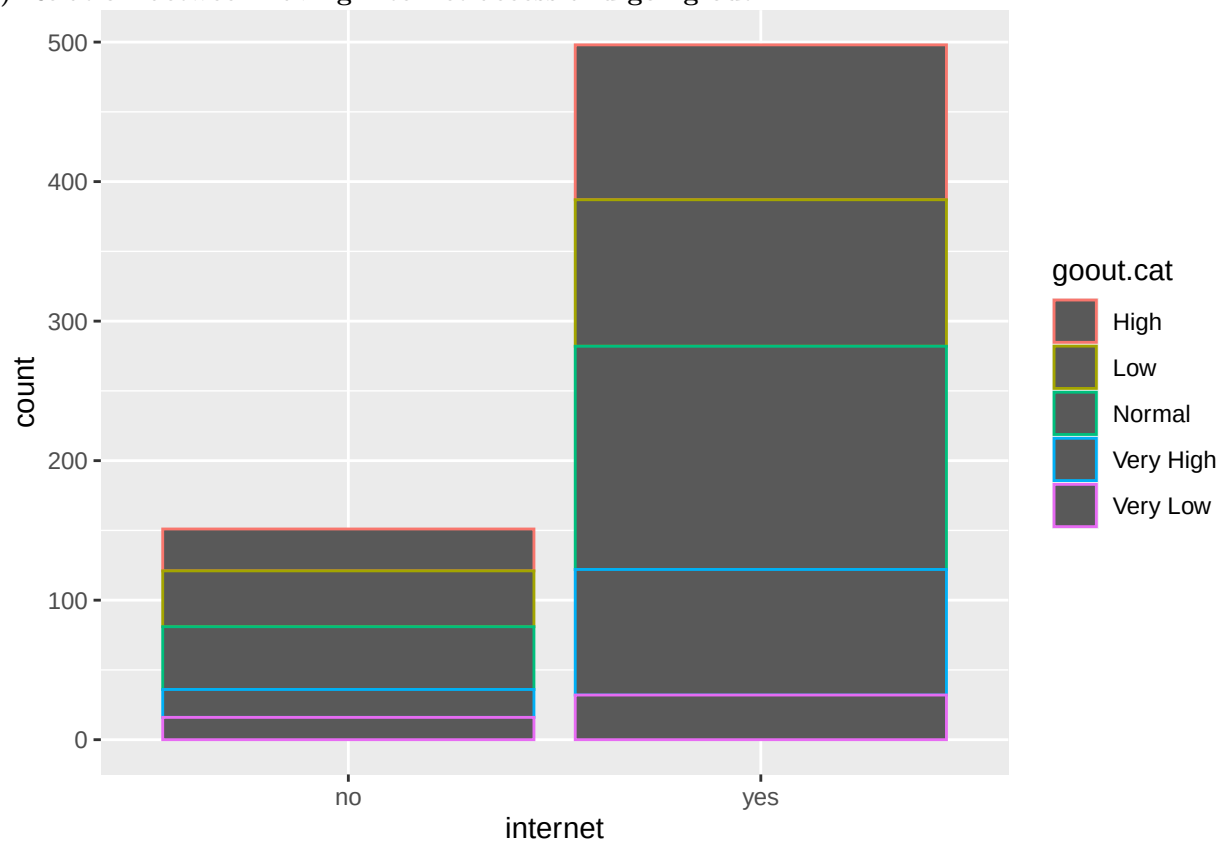
3) Relation between performance(failures in past) and absences



From the visualization, we can say that majority of the students with 0 absences have 0 failures but is inconclusive for other data.

```
ggplot(data = student, mapping = aes(x=internet, colour=goout.cat)) +  
  geom_bar()
```

4) Relation between having internet access and going out



From the visualization, we can say that the availability of internet is not a parameter for students going out with friends.